



ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT106-3-2-SNA

SYSTEMS AND NETWORK ADMINISTRATION

APD2F2411CS(CYB)

HAND OUT DATE: WEEK 2 NOVEMBER 2024

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WEIGHTAGE: 30%

INSTRUCTIONS TO CANDIDATES:

- 1 Submit your assignment at the administrative counter.**
- 2 Students are advised to underpin their answers with the use of references (cited using the Harvard Name System of Referencing).**
- 3 Late submission will be awarded zero (0) unless Extenuating Circumstances (EC) are upheld.**
- 4 Cases of plagiarism will be penalized.**
- 5 The assignment should be bound in an appropriate style (comb bound or stapled).**
- 6 Where the assignment should be submitted in both hardcopy and softcopy, the softcopy of the written assignment and source code (where appropriate) should be on a CD in an envelope / CD cover and attached to the hardcopy.**
- 7 You must obtain 50% overall to pass this module.**

Version 3 EC 2019-May



INDIVIDUAL ASSIGNMENT

SYSTEM AND NETWORK ADMINISTRATION

CT106-3-2-SNA

STUDENT NAME: HEMANTHRAJAH A/L N KRISHNARAJAH

TP NO: TP070844

INTAKE CODE: APD2F2411CS(CYB)

MODULE CODE: CT106-3-2-SNA

PROJECT TITLE: Individual SNA Network Administration Project

**LECTURER NAME: TS. MOHAMAD ASRI BIN AHMAD, SHAHAB
ALIZADEH**

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1.0 Introduction

My assignment focuses on building and setting up a completely functional email server using Rocky Linux. I have chosen Oracle VirtualBox as my virtualized environment, which allows me to construct an isolated and safe testing setup. In this project, I configured the server with the FQDN hemanthserver.bungkus.net and the client as hemanthclient.bungkus.net, enabling them to connect within a dedicated network.

To accomplish this project, I installed and configured Postfix, a Mail Transfer Agent (MTA), to handle sending and receiving emails. I also utilized Dovecot as the IMAP and POP3 server, which allows users to get their emails and manages authentication securely. I ensured that the entire communication system was secured using SSL/TLS encryption, which prevents sensitive information from being exposed during transmission.

In addition to installing the main email server components, I worked on setting up DNS, static IPs, and user authentication techniques. To test my configuration, I used Thunderbird as the email client to ensure that emails could be sent and received successfully between the server and the client.

This assignment offered me a solid grasp of how to set up, configure, and debug an email server. By recording every stage of my approach, I've built a resource that I can refer to in the future or use as a guide for similar setups. It also underlined the necessity of security and precision in establishing modern communication networks. This endeavor has been a fantastic learning experience for me.

1.1 introduction to Virtual Machines

Virtual machines (VMs) are software-based systems that imitate the capabilities of a physical computer. They allow users to run several operating systems on a single physical machine by establishing separate settings called virtual environments. Virtual machines rely on hypervisors to control the hardware resources, offering flexibility, cost effectiveness, and ease of testing. For example, I can utilize VMs to experiment with multiple operating systems, configurations, and software without affecting my host machine. VMs are extremely handy for network and server-related projects since they allow me to establish many systems that interact with each other in a controlled environment.



Figure 1: VirtualBox Logo

For this assignment, I utilized Oracle VirtualBox as my virtualization program. It is an open-source program that helps me construct and manage virtual machines simply. I chose VirtualBox because it is lightweight, user-friendly, and supports a broad variety of operating systems. My virtual environment comprised of two virtual machines: one running Rocky Linux as the server and the other running Ubuntu as the client.



Figure 2: Rocky Server Interface

Rocky Linux was my pick for the server because it is a stable, enterprise-grade operating system intended for managing server chores. It is a downstream version of Red Hat Enterprise Linux (RHEL), which means it's dependable, secure, and excellent for hosting services like email servers. I configured Rocky Linux with the FQDN `hemanthserver.bungkus.net`.

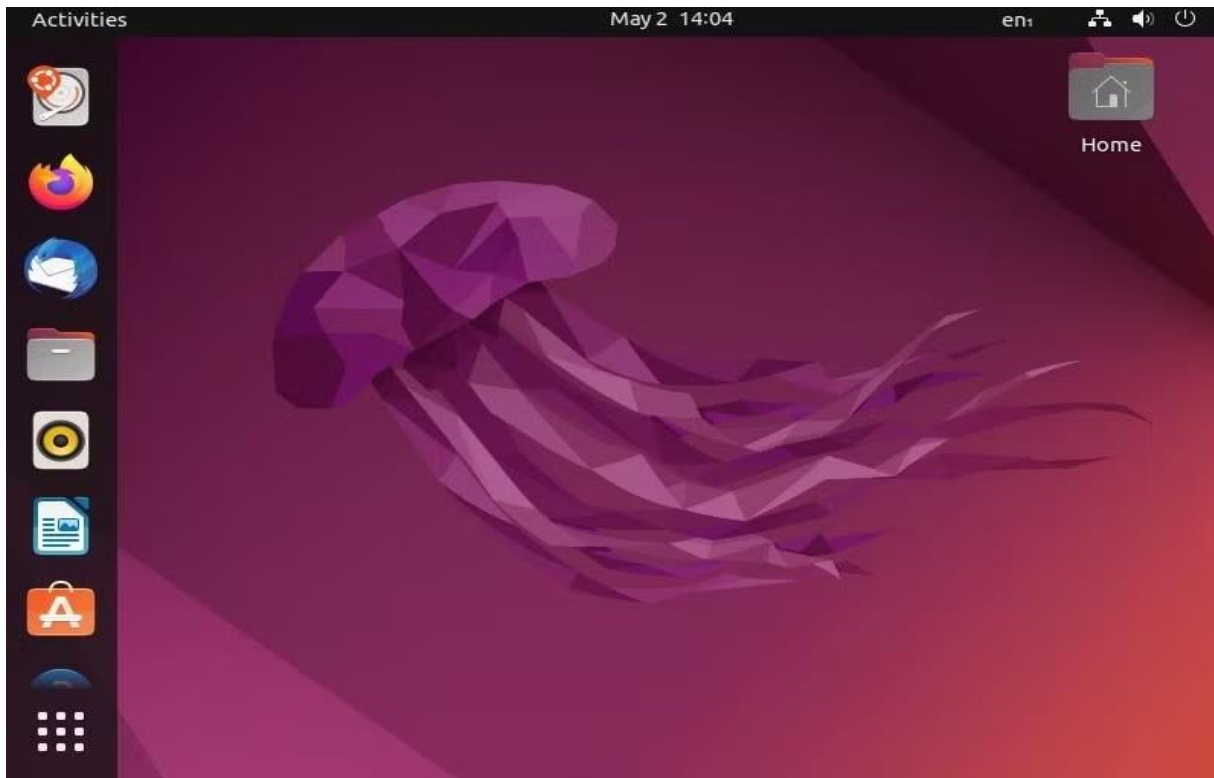


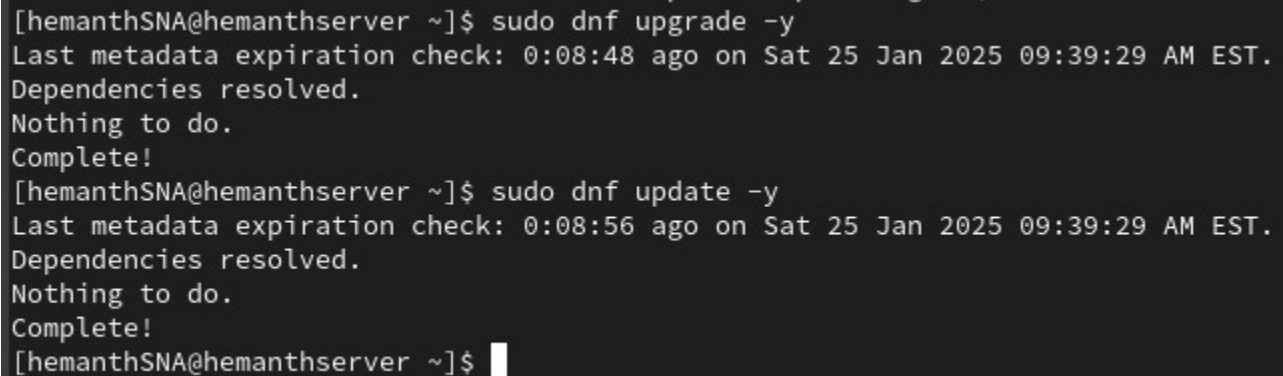
Figure 3: Ubuntu Interface

On the client side, I selected Ubuntu, a famous Linux system noted for its simplicity and usefulness. Ubuntu supplied me with an easy interface and a strong collection of tools to communicate with the server. I configured Ubuntu with the FQDN `hemanthclient.bungkus.net` to communicate efficiently with the server.

This configuration enables me to emulate a real-world email server and client scenario. It gave me the opportunity to examine the functionality of virtual machines while setting up and testing server-client communication.

2.0 Installation and Configuration of Postfix (MTA)

Setting up the email server, I started by installing and configuring Postfix, a sophisticated and widely-used Mail Transfer Agent (MTA). Postfix is responsible for sending, receiving, and relaying emails, making it an essential component of my email server. Its modular design, robust security measures, and efficiency make it the right choice for our project. Below, I will explain each step I completed during the installation and configuration process, along with extensive explanations of the commands and parameters used (Codex, 2024).

A terminal window with a dark background and light-colored text. The prompt is [hemanthSNA@hemanthserver ~]\$. The first command is sudo dnf upgrade -y. The output shows the last metadata expiration check, dependencies resolved, nothing to do, and complete. The second command is sudo dnf update -y. The output is similar, showing the last metadata expiration check, dependencies resolved, nothing to do, and complete. The prompt returns to [hemanthSNA@hemanthserver ~]\$.

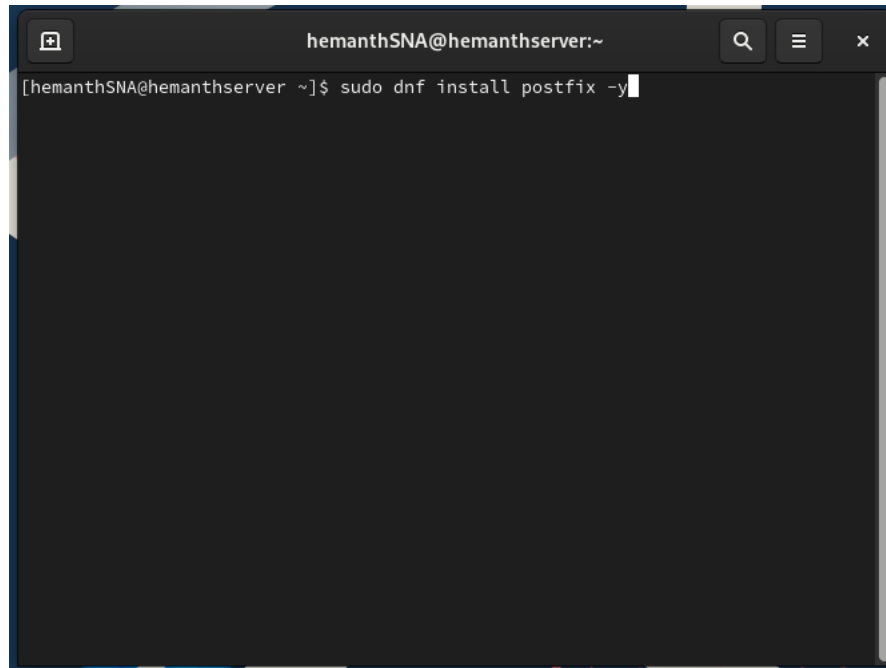
```
[hemanthSNA@hemanthserver ~]$ sudo dnf upgrade -y
Last metadata expiration check: 0:08:48 ago on Sat 25 Jan 2025 09:39:29 AM EST.
Dependencies resolved.
Nothing to do.
Complete!
[hemanthSNA@hemanthserver ~]$ sudo dnf update -y
Last metadata expiration check: 0:08:56 ago on Sat 25 Jan 2025 09:39:29 AM EST.
Dependencies resolved.
Nothing to do.
Complete!
[hemanthSNA@hemanthserver ~]$
```

Figure 4: Update & Upgrade Rocky Linux

First and foremost, we should always update and upgrade our system to make sure we are running the latest patch and the latest packages to ensure smooth installations of any software or application we're trying to install.

2.1 Installation of Postfix (MTA)

To install Postfix on my Rocky Linux virtual machine, I ran the following command:

A terminal window with a dark background. The title bar shows 'hemanthSNA@hemanthserver:~'. The prompt is '[hemanthSNA@hemanthserver ~]\$' and the command entered is 'sudo dnf install postfix -y'. The cursor is at the end of the command.

```
hemanthSNA@hemanthserver:~  
[hemanthSNA@hemanthserver ~]$ sudo dnf install postfix -y
```

Figure 5: Install Postfix

- **sudo:** This command grants me administrative privileges to perform the installation, as modifying system software requires root access.
- **dnf install postfix:** This part instructs the package manager to fetch and install Postfix along with its necessary dependencies.
- **-y:** By including this flag, I automatically confirm the installation process, avoiding the need to manually approve prompts during installation.

```
[hemanthSNA@hemanthserver ~]$ sudo dnf install postfix -y
[sudo] password for hemanthSNA:
Rocky Linux 9 - BaseOS                1.8 kB/s | 4.1 kB    00:02
Rocky Linux 9 - AppStream              5.9 kB/s | 4.5 kB    00:00
Rocky Linux 9 - AppStream              3.7 MB/s | 8.5 MB    00:02
Rocky Linux 9 - Extras                 1.4 kB/s | 2.9 kB    00:02
Dependencies resolved.
=====
Package            Architecture Version           Repository      Size
=====
Installing:
postfix            x86_64        2:3.5.25-1.el9   appstream      1.5 M
=====
Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
Downloading Packages:
postfix-3.5.25-1.el9.x86_64.rpm        3.1 MB/s | 1.5 MB    00:00
-----
Total                                  1.2 MB/s | 1.5 MB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Running scriptlet: postfix-2:3.5.25-1.el9.x86_64 : 1/1
  Installing           : postfix-2:3.5.25-1.el9.x86_64 : 1/1
  Running scriptlet: postfix-2:3.5.25-1.el9.x86_64 : 1/1
  Verifying            : postfix-2:3.5.25-1.el9.x86_64 : 1/1

Installed:
  postfix-2:3.5.25-1.el9.x86_64

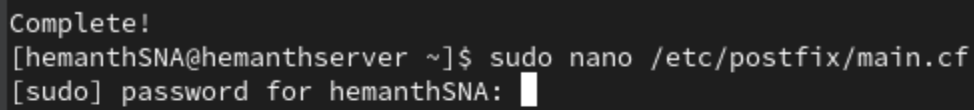
Complete!
[hemanthSNA@hemanthserver ~]$
```

Figure 6: Complete Installation of Postfix

This command ensured that Postfix was successfully downloaded and installed on my system, setting the stage for further configuration.

2.2 Configuration of Postfix (MTA)

After the installation, I proceeded to configure Postfix to match the specific requirements of my setup (trans ip, n.d.). The main configuration file for Postfix is `/etc/postfix/main.cf`, where most of the server's behavior is defined. I opened this file for editing using:

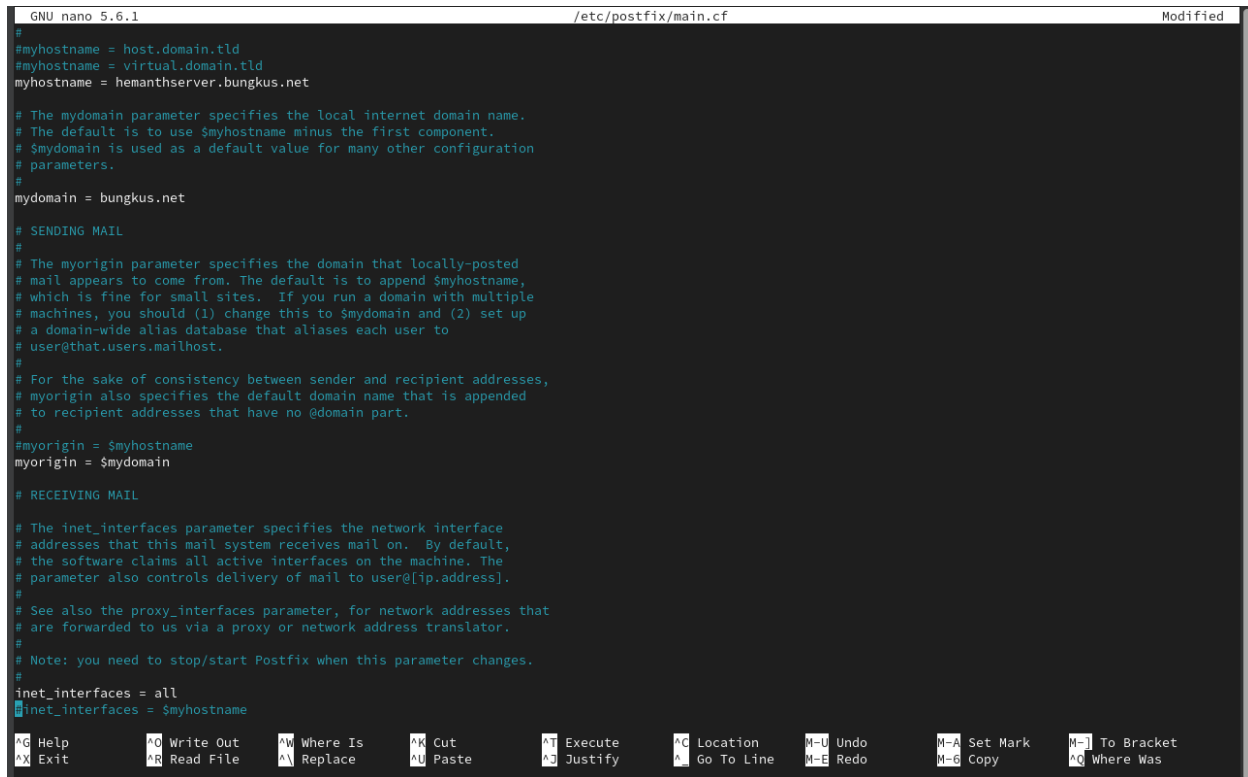
A terminal window with a dark background. The text shows the completion of an installation, followed by a user running the 'sudo nano /etc/postfix/main.cf' command. The prompt then changes to '[sudo] password for hemanthSNA:' with a cursor at the end.

```
Complete!  
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/postfix/main.cf  
[sudo] password for hemanthSNA: █
```

Figure 7: `sudo nano /etc/postfix/main.cf`

The `sudo` command ensured I had the necessary permissions, while `nano` is a simple text editor I used to make changes to the configuration.

2.2.1 Setting the Hostname, Defining the Domain, Setting the Origin, Configuring Network Interfaces



```
GNU nano 5.6.1 /etc/postfix/main.cf Modified
#
#myhostname = host.domain.tld
#myhostname = virtual.domain.tld
myhostname = hemanthserver.bungkus.net

# The mydomain parameter specifies the local internet domain name.
# The default is to use $myhostname minus the first component.
# $mydomain is used as a default value for many other configuration
# parameters.
#
mydomain = bungkus.net

# SENDING MAIL
#
# The myorigin parameter specifies the domain that locally-posted
# mail appears to come from. The default is to append $myhostname,
# which is fine for small sites. If you run a domain with multiple
# machines, you should (1) change this to $mydomain and (2) set up
# a domain-wide alias database that aliases each user to
# user@that.users.mailhost.
#
# For the sake of consistency between sender and recipient addresses,
# myorigin also specifies the default domain name that is appended
# to recipient addresses that have no @domain part.
#
#myorigin = $myhostname
myorigin = $mydomain

# RECEIVING MAIL
#
# The inet_interfaces parameter specifies the network interface
# addresses that this mail system receives mail on. By default,
# the software claims all active interfaces on the machine. The
# parameter also controls delivery of mail to user@[ip.address].
#
# See also the proxy_interfaces parameter, for network addresses that
# are forwarded to us via a proxy or network address translator.
#
# Note: you need to stop/start Postfix when this parameter changes.
#
inet_interfaces = all
inet_interfaces = $myhostname

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^_ Set Mark   ^= To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^D Justify    ^G Go To Line ^E Redo       ^M Copy      ^Q Where Was
```

Figure 8: Configuration of Postfix

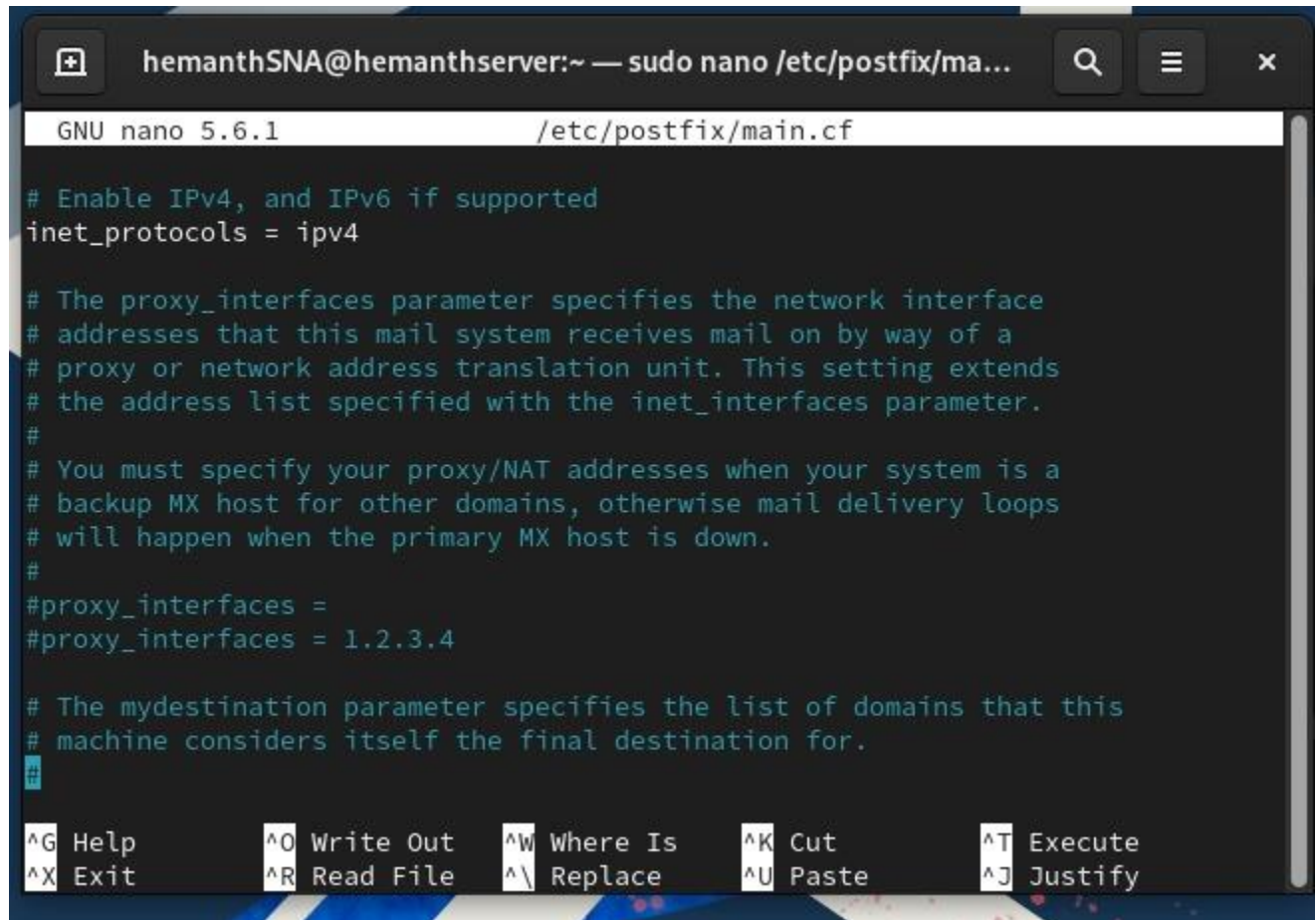
The **myhostname** parameter specifies the Fully Qualified Domain Name (FQDN) of the mail server. This is essential because it identifies the server on my network and is used during email exchanges to validate the server's identity. In my setup, I set it to `hemanthserver.bungkus.net`, which is the server's FQDN.

The **mydomain** parameter sets the domain name for the server. This is the portion of the email address that comes after the `@` symbol (e.g., `user@bungkus.net`). This configuration ensures that emails sent from my server will be associated with my `bungkus.net` domain.

The **myorigin** parameter defines what domain should be appended to locally sent emails. By setting it to \$mydomain, I ensured that emails sent from system processes or local accounts would appear to come from @bungkus.net. This provides a consistent and professional appearance for all outgoing emails.

The **inet_interfaces** parameter specifies which network interfaces Postfix should listen on. By setting it to all, I ensured that Postfix would accept connections from all available interfaces, including localhost and external networks. This is critical for allowing communication between the client and server in my setup.

2.2.2 Defining Protocols



```
hemanthSNA@hemanthserver:~ — sudo nano /etc/postfix/main.cf
GNU nano 5.6.1 /etc/postfix/main.cf

# Enable IPv4, and IPv6 if supported
inet_protocols = ipv4

# The proxy_interfaces parameter specifies the network interface
# addresses that this mail system receives mail on by way of a
# proxy or network address translation unit. This setting extends
# the address list specified with the inet_interfaces parameter.
#
# You must specify your proxy/NAT addresses when your system is a
# backup MX host for other domains, otherwise mail delivery loops
# will happen when the primary MX host is down.
#
#proxy_interfaces =
#proxy_interfaces = 1.2.3.4

# The mydestination parameter specifies the list of domains that this
# machine considers itself the final destination for.
#
```

Figure 9: Configuration of Postfix cont.

The **inet_protocols** parameter defines the network protocols Postfix should support. In my case, I specified `ipv4` because my network is configured to use IPv4. This prevents Postfix from attempting to use IPv6, which might not be supported in certain environments.

2.2.3 Specifying Accepted Domains

```

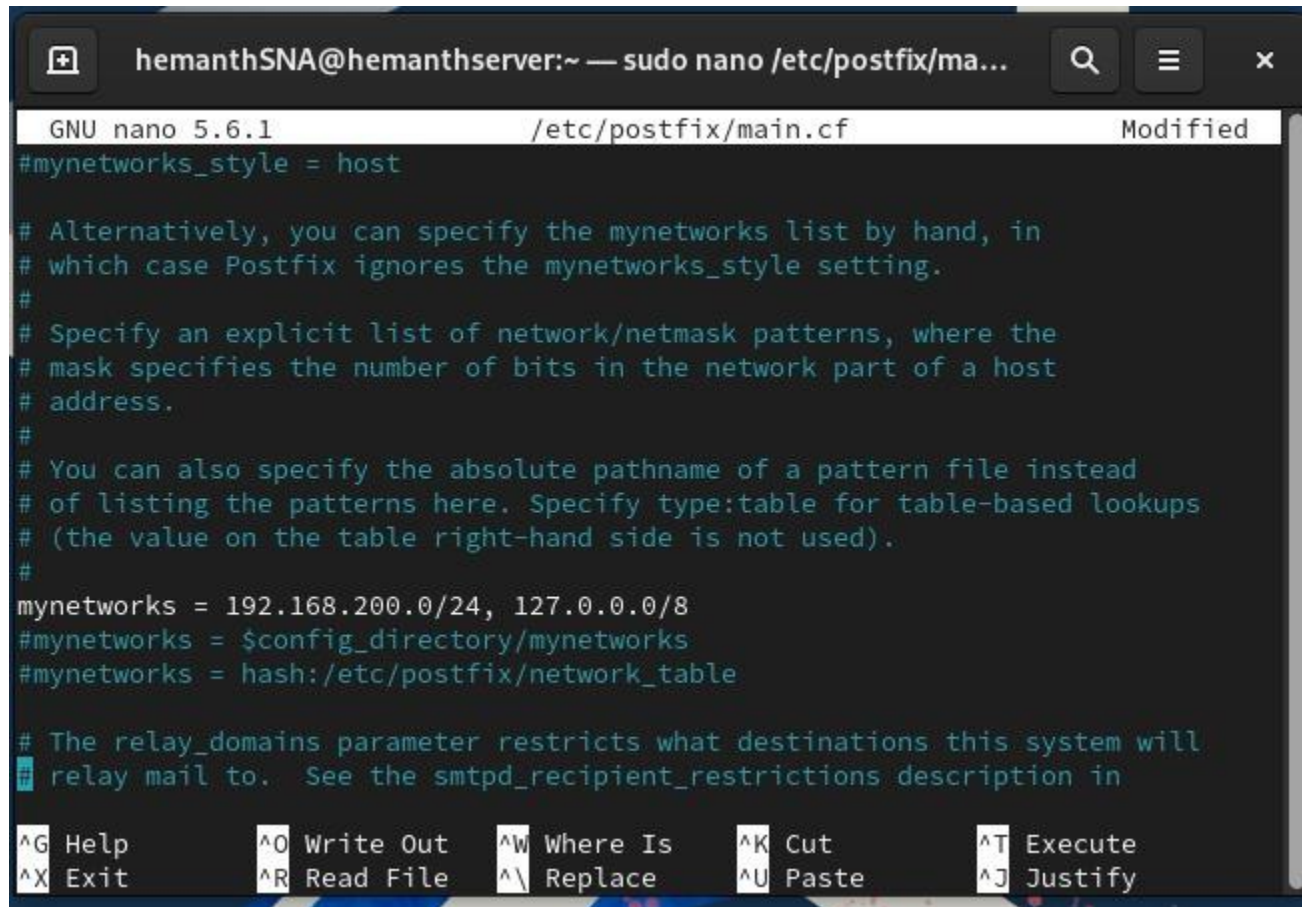
GNU nano 5.6.1 /etc/postfix/main.cf Modified
# the SMTP server, or use permit_mx_backup if you are lazy (see
# STANDARD_CONFIGURATION_README).
#
# The local machine is always the final destination for mail addressed
# to user@[the.net.work.address] of an interface that the mail system
# receives mail on (see the inet_interfaces parameter).
#
# Specify a list of host or domain names, /file/name or type:table
# patterns, separated by commas and/or whitespace. A /file/name
# pattern is replaced by its contents; a type:table is matched when
# a name matches a lookup key (the right-hand side is ignored).
# Continue long lines by starting the next line with whitespace.
#
# See also below, section "REJECTING MAIL FOR UNKNOWN LOCAL USERS".
#
#mydestination = $myhostname, localhost.$mydomain, localhost
#mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain,
               mail.$mydomain, www.$mydomain, ftp.$mydomain
#
# REJECTING MAIL FOR UNKNOWN LOCAL USERS
#
# The local_recipient_maps parameter specifies optional lookup tables
# with all names or addresses of users that are local with respect
# to $mydestination, $inet_interfaces or $proxy_interfaces.
#
# If this parameter is defined, then the SMTP server will reject
# mail for unknown local users. This parameter is defined by default.
#
# To turn off local recipient checking in the SMTP server, specify
# local_recipient_maps = (i.e. empty).
#
# The default setting assumes that you use the default Postfix local
# delivery agent for local delivery. You need to update the
# local_recipient_maps setting if:
#
# - You define $mydestination domain recipients in files other than
#   /etc/passwd, /etc/aliases, or the $virtual_alias_maps files.
#   For example, you define $mydestination domain recipients in
#   the $virtual_mailbox_maps files.
#
# - You redefine the local delivery agent in master.cf.
^C Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark   M-I To Bracket
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-B Copy      ^Q Where Was

```

Figure 10: Configuration of Postfix cont.

The **mydestination** parameter lists all the domains the server considers local. Emails addressed to these domains will be delivered locally rather than being forwarded elsewhere. This ensures that Postfix handles mail for hemanthserver.bungkus.net, localhost, and bungkus.net.

2.2.4 Setting Trusted Networks



```
hemanthSNA@hemanthserver:~ — sudo nano /etc/postfix/main.cf
GNU nano 5.6.1 /etc/postfix/main.cf Modified
#mynetworks_style = host

# Alternatively, you can specify the mynetworks list by hand, in
# which case Postfix ignores the mynetworks_style setting.
#
# Specify an explicit list of network/netmask patterns, where the
# mask specifies the number of bits in the network part of a host
# address.
#
# You can also specify the absolute pathname of a pattern file instead
# of listing the patterns here. Specify type:table for table-based lookups
# (the value on the table right-hand side is not used).
#
mynetworks = 192.168.200.0/24, 127.0.0.0/8
#mynetworks = $config_directory/mynetworks
#mynetworks = hash:/etc/postfix/network_table

# The relay_domains parameter restricts what destinations this system will
# relay mail to. See the smtpd_recipient_restrictions description in

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Figure 11: Configuration of Postfix cont.

The **mynetworks** parameter specifies the trusted networks that can send emails through the server without authentication. I included the local network 192.168.200.0/24 and the loopback address 127.0.0.0/8. This configuration allows my client machine (hemanthclient.bungkus.net) and localhost to relay mail without restrictions.

2.2.5 Configuring Mailbox Format

```

GNU nano 5.6.1 /etc/postfix/main.cf Modified
# mailbox file relative to a user's home directory. The default
# mailbox file is /var/spool/mail/user or /var/mail/user. Specify
# "Maildir/" for qmail-style delivery (the / is required).
#
#home_mailbox = Mailbox
home_mailbox = Maildir/

# The mail_spool_directory parameter specifies the directory where
# UNIX-style mailboxes are kept. The default setting depends on the
# system type.
#
#mail_spool_directory = /var/mail
#mail_spool_directory = /var/spool/mail

# The mailbox_command parameter specifies the optional external
# command to use instead of mailbox delivery. The command is run as
# the recipient with proper HOME, SHELL and LOGNAME environment settings.
# Exception: delivery for root is done as $default_user.
#
# Other environment variables of interest: USER (recipient username),
# EXTENSION (address extension), DOMAIN (domain part of address),
# and LOCAL (the address localpart).
#
# Unlike other Postfix configuration parameters, the mailbox_command
# parameter is not subjected to $parameter substitutions. This is to
# make it easier to specify shell syntax (see example below).
#
# Avoid shell meta characters because they will force Postfix to run
# an expensive shell process. Procmail alone is expensive enough.
#
# IF YOU USE THIS TO DELIVER MAIL SYSTEM-WIDE, YOU MUST SET UP AN
# ALIAS THAT FORWARDS MAIL FOR ROOT TO A REAL USER.
#
#mailbox_command = /some/where/procmail
#mailbox_command = /some/where/procmail -a "$EXTENSION"
mailbox_command =

# The mailbox_transport specifies the optional transport in master.cf
# to use after processing aliases and .forward files. This parameter
# has precedence over the mailbox_command, fallback_transport and
# luser_relay parameters.
#
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-J To Bracket
^X Exit      ^R Read File  ^\ Replace    ^P Paste      ^D Justify    ^_ Go To Line  M-E Redo      M-G Copy      ^Q Where Was

```

Figure 12: Configuration of Postfix cont.

The **home_mailbox** parameter defines the directory where user emails will be stored. By setting it to **Maildir/**, I chose the **Maildir** format, which stores each email as a separate file. This format is widely used for its compatibility with various email clients and ease of access.

2.2.6 Enabling Secure Connections

```

GNU nano 5.6.1 /etc/postfix/main.cf Modified
# Basic Postfix TLS configuration by default with self-signed certificate
# for inbound SMTP and also opportunistic TLS for outbound SMTP.

# The full pathname of a file with the Postfix SMTP server RSA certificate
# in PEM format. Intermediate certificates should be included in general,
# the server certificate first, then the issuing CA(s) (bottom-up order).
#
smtpd_tls_cert_file = /etc/ssl/certs/mailcert.pem

# The full pathname of a file with the Postfix SMTP server RSA private key
# in PEM format. The private key must be accessible without a pass-phrase,
# i.e. it must not be encrypted.
#
smtpd_tls_key_file = /etc/ssl/private/mailkey.pem

# Announce STARTTLS support to remote SMTP clients, but do not require that
# clients use TLS encryption (opportunistic TLS inbound).
#
smtpd_tls_security_level = may

# Directory with PEM format Certification Authority certificates that the
# Postfix SMTP client uses to verify a remote SMTP server certificate.
#
smtp_tls_CApath = /etc/pki/tls/certs

# The full pathname of a file containing CA certificates of root CAs
# trusted to sign either remote SMTP server certificates or intermediate CA
# certificates.
#
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt

# Use TLS if this is supported by the remote SMTP server, otherwise use
# plaintext (opportunistic TLS outbound).
#
smtp_tls_security_level = may
meta_directory = /etc/postfix
shlib_directory = /usr/lib64/postfix

smtpd_use_tls = yes
smtpd_tls_session_cache_database = btree:${data_directory}/smtpd_scache
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
  
```

Figure 13: Configuration of Postfix cont.

General Configuration:

1. **meta_directory = /etc/postfix**

1.1. This line defines the directory where Postfix stores its configuration files. It's essential for Postfix to function correctly.

2. **shlib_directory = /usr/lib64/postfix**

2.1. This line specifies the directory where Postfix's shared libraries are located. These libraries are necessary for various Postfix components.

TLS Configuration:**3. smtpd_use_tls = yes**

3.1. This is the primary configuration for enabling TLS encryption on the server side. By setting it to "yes," Postfix will attempt to establish encrypted connections with clients.

4. smtpd_tls_cert_file = /etc/ssl/certs/mailcert.pem

4.1. This line specifies the path to the SSL certificate file that Postfix will use to present its identity to clients. The certificate must be in PEM format.

5. smtpd_tls_key_file = /etc/ssl/private/mailkey.pem

5.1. This line specifies the path to the private key file that corresponds to the certificate. The private key must also be in PEM format and **should not be passphrase-protected**.

6. smtpd_tls_security_level = may

6.1. This setting controls how strictly TLS encryption is enforced. "may" means that Postfix will offer TLS encryption to clients but will still allow unencrypted connections if the client doesn't support TLS.

7. smtp_tls_CApath = /etc/pki/tls/certs

7.1. This line specifies the directory where Postfix can find Certificate Authority (CA) certificates used to verify the authenticity of the certificates presented by remote SMTP servers.

8. smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt

8.1. This line specifies the file containing CA certificates used for certificate verification. This is an alternative to using the smtp_tls_CApath setting.

9. smtp_tls_security_level = may

9.1. Same as smtpd_tls_security_level, this controls the level of TLS encryption enforcement when Postfix is acting as a client (e.g., sending mail to other servers).

```
# Use TLS if this is supported by the remote SMTP server, otherwise use
# plaintext (opportunistic TLS outbound).
#
smtpd_recipient_restrictions = permit_mynetworks, permit_sasl_authenticated, reject_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = cyrus
smtpd_sasl_path = smtpd
smtpd_sasl_security_options = noanonymous
smtpd_tls_auth_only = yes
broken_sasl_auth_clients = yes
meta_directory = /etc/postfix
shlib_directory = /usr/lib64/postfix

smtpd_use_tls = yes
smtpd_tls_session_cache_database = btree:${data_directory}/smtpd_scache
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
```

Figure 14: Configuration of Postfix cont.

Recipient Restrictions:

1. **smtpd_recipient_restrictions = permit_mynetworks, permit_sasl_authenticated, reject_unauth_destination¹**

1.1. This line defines the rules for determining whether to accept or reject mail for a particular recipient.

1.1.1. **permit_mynetworks**: Allows mail to be delivered to recipients within the local network (defined by mynetworks).

1.1.2. **permit_sasl_authenticated**: Allows mail to be delivered to recipients if the sender has successfully authenticated using SASL (Simple Authentication and Security Layer).

1.1.3. **reject_unauth_destination**: Rejects mail for destinations that are not explicitly allowed by other rules.

SASL (Simple Authentication and Security Layer) Configuration:

2. **smtpd_sasl_auth_enable = yes**

2.1. Enables SASL authentication for incoming mail.

3. smtpd_sasl_type = cyrus

3.1. Specifies that Cyrus SASL is used for authentication.

4. smtpd_sasl_path = smtpd

4.1. Specifies the path to the SASL library.

5. smtpd_sasl_security_options = noanonymous

5.1. Disables anonymous authentication.

6. smtpd_tls_auth_only = yes

6.1. Requires that TLS encryption is used for all authenticated connections.

7. broken_sasl_auth_clients = yes

7.1. Enables support for clients that may not implement SASL correctly.

2.2.7 Restarting Postfix and Enable Postfix

```
[hemanthSNA@hemanthserver ~]$ sudo systemctl restart postfix
[sudo] password for hemanthSNA:
[hemanthSNA@hemanthserver ~]$ sudo systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
[hemanthSNA@hemanthserver ~]$
```

Figure 15: Configuration of Postfix cont.

The `systemctl restart` and `enable postfix` command ensures that the service reloads with the updated settings and gets started if it has not been already.

Installing and configuring Postfix was a critical step in setting up my email server. By carefully editing the `main.cf` file and tailoring it to my specific needs, I ensured that the server could handle incoming and outgoing mail efficiently and securely. This process also gave me valuable insight into how email servers function and the role Postfix plays in modern communication systems.

3.0 Installation and Configuration of Dovecot

3.1 Installation of Dovecot

After setting up Postfix for sending and receiving emails, I continued to install and configure Dovecot, a sophisticated IMAP and POP3 server (Postfix Email Server, n.d.). Dovecot complements Postfix by letting users access and control their emails through email clients like Thunderbird. It also includes key functionality such as user authentication, mail retrieval, and secure communication. Below, I outline the procedures I followed during the installation and configuration of Dovecot, along with explanations for each operation.

To begin, I installed Dovecot using the package manager. The command I used was:

```
[hemanthSNA@hemanthserver ~]$ sudo dnf install dovecot -y
Last metadata expiration check: 0:36:20 ago on Thu 23 Jan 2025 10:58:17 AM EST.
Dependencies resolved.
=====
Package                                Architecture      Version                                Repository          Size
=====
Installing:
dovecot                                x86_64            1:2.3.16-14.el9                      appstream            4.7 M
Installing dependencies:
clucene-core                           x86_64            2.3.3.4-42.20130812.e8e3d20git.el9    appstream            585 k
libexttextcat                           x86_64            3.4.5-11.el9                          appstream            209 k
=====
Transaction Summary
=====
Install 3 Packages

Total download size: 5.5 M
Installed size: 20 M
Downloading Packages:
(1/3): libexttextcat-3.4.5-11.el9.x86_64.rpm                                301 kB/s | 209 kB  00:00
(2/3): clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64.rpm          761 kB/s | 585 kB  00:00
(3/3): dovecot-2.3.16-14.el9.x86_64.rpm                                    1.6 MB/s | 4.7 MB  00:02
-----
Total                                                                    1.1 MB/s | 5.5 MB  00:04
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Installing     : libexttextcat-3.4.5-11.el9.x86_64                1/3
  Installing     : clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64 2/3
  Running scriptlet: dovecot-1:2.3.16-14.el9.x86_64                3/3
```

Figure 16: Install Dovecot

sudo: This command ensures I have the necessary administrative privileges to install software packages.

dnf install dovecot: Instructs the package manager to download and install Dovecot along with its dependencies.

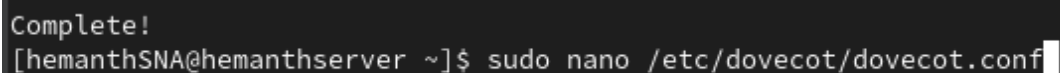
-y: Automatically confirms the installation without requiring manual input.

This command ensured that Dovecot was installed and ready for configuration on my server. These configurations are similar to installing Postfix.

3.2 Configuration of Dovecot

Dovecot's configuration files are located in `/etc/dovecot/conf.d/`, and its primary configuration file is `/etc/dovecot/dovecot.conf`. The files in `/etc/dovecot/conf.d/` are modular, meaning each file is dedicated to a specific aspect of Dovecot's functionality, such as authentication, SSL, and mail location. Editing these files allows for a highly customizable setup.

To begin configuring Dovecot, I opened its main configuration file:

A terminal window with a dark background. The first line shows the word "Complete!". The second line shows the command prompt "[hemanthSNA@hemanthserver ~]\$ sudo nano /etc/dovecot/dovecot.conf" followed by a white cursor block.

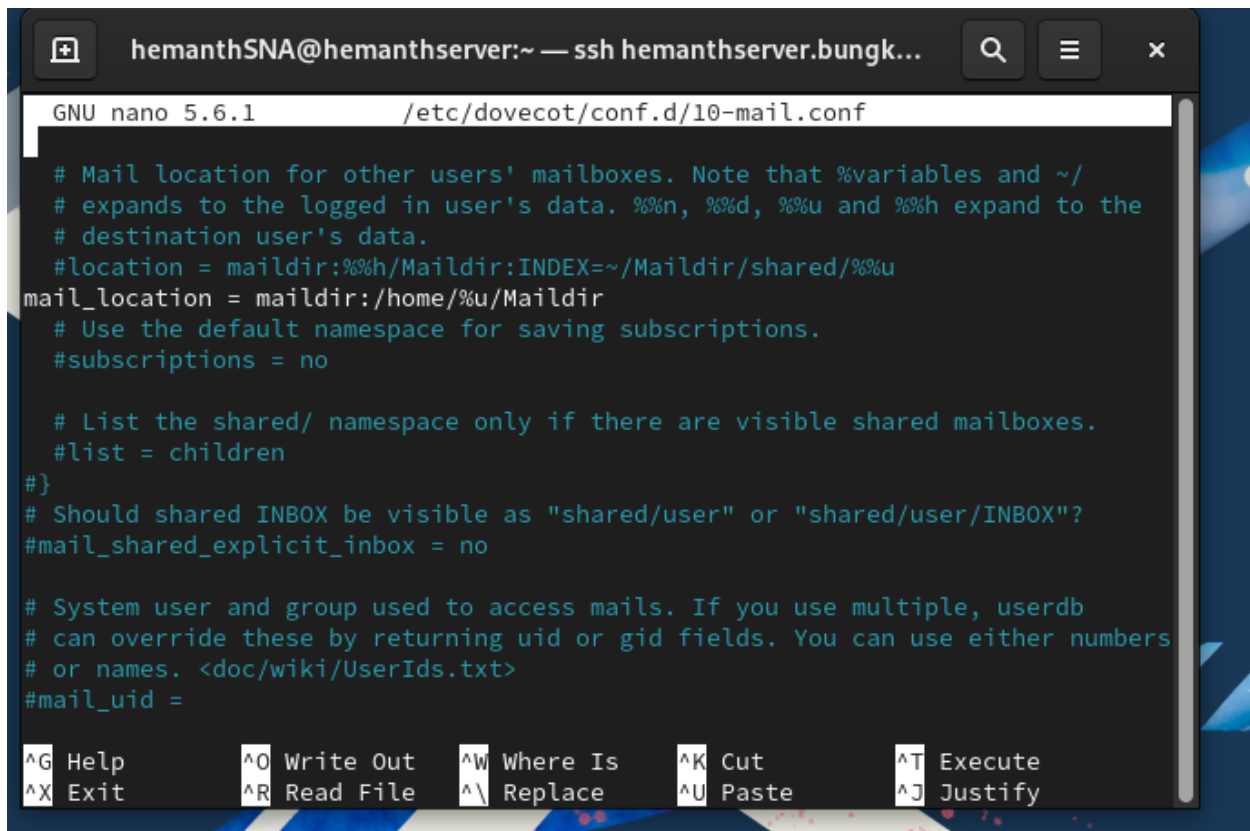
```
Complete!  
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/dovecot.conf
```

Figure 17: `sudo nano /etc/dovecot/dovecot.conf`

Using this command, I could ensure Dovecot's overall settings matched the requirements of my email server.

3.2.1 Configuring Mail Location

The mail location determines where user emails are stored on the server (Codex, Configuring Postfix Mail Server on Ubuntu 20.04, 2024). I configured this setting in `/etc/dovecot/conf.d/10-mail.conf`:



```
hemanthSNA@hemanthserver:~ — ssh hemanthserver.bungk...
GNU nano 5.6.1 /etc/dovecot/conf.d/10-mail.conf

# Mail location for other users' mailboxes. Note that %variables and ~/
# expands to the logged in user's data. %%n, %%d, %%u and %%h expand to the
# destination user's data.
#location = maildir:%%h/Maildir:INDEX=~ /Maildir/shared/%%u
mail_location = maildir:/home/%u/Maildir
# Use the default namespace for saving subscriptions.
#subscriptions = no

# List the shared/ namespace only if there are visible shared mailboxes.
#list = children
#}
# Should shared INBOX be visible as "shared/user" or "shared/user/INBOX"?
#mail_shared_explicit_inbox = no

# System user and group used to access mails. If you use multiple, userdb
# can override these by returning uid or gid fields. You can use either numbers
# or names. <doc/wiki/UserIds.txt>
#mail_uid =

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Figure 18: Mail location

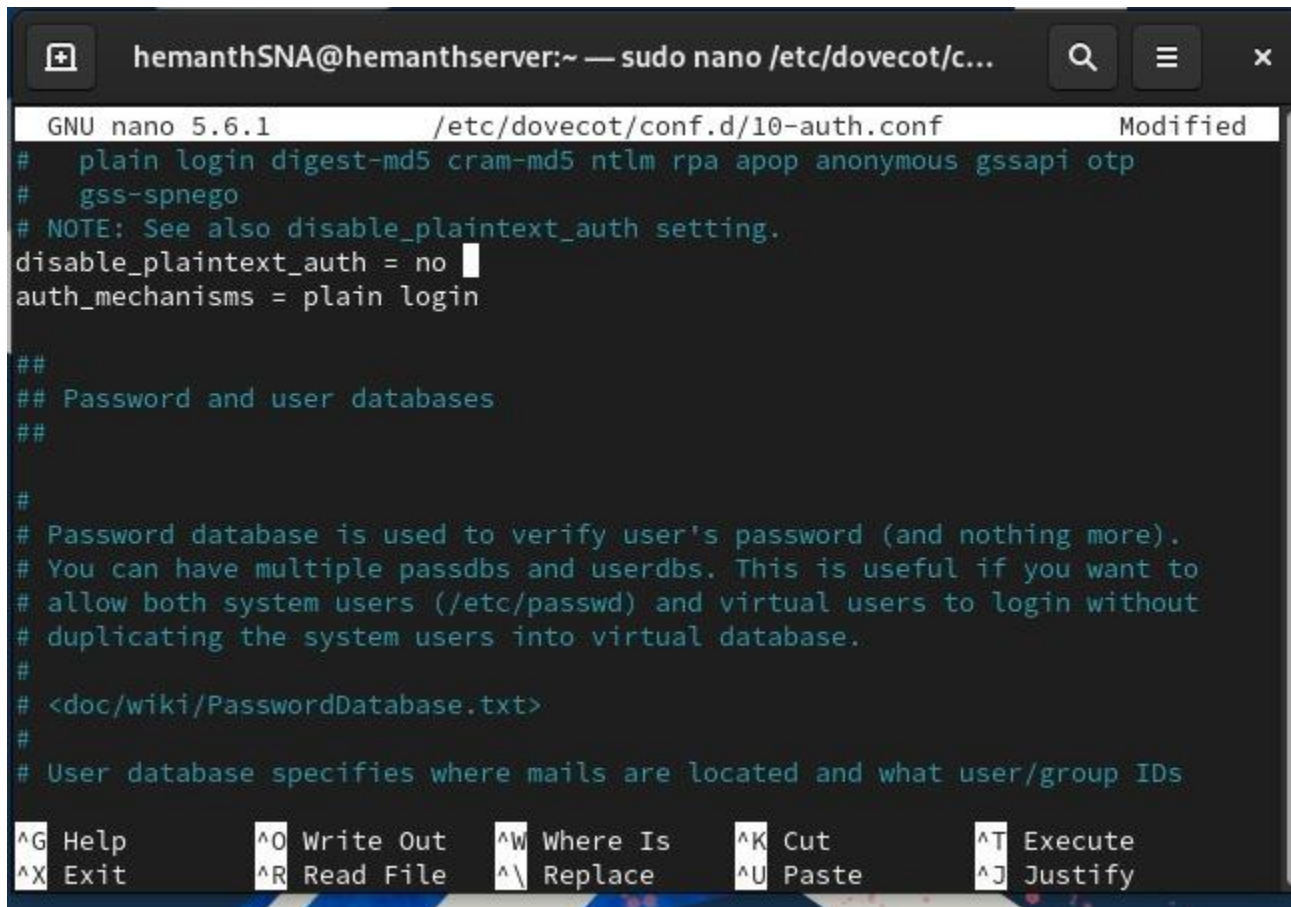
mail_location: Specifies the directory structure where emails are stored.

maildir: Indicates that emails should be stored in the Maildir format, which is widely used for its reliability and compatibility with most email clients.

/home/%u/Maildir: Defines the directory structure, where %u represents the domain (e.g., bungkus.net).

3.2.2 Configuring Authentication

To allow users to log in and access their mailboxes, I configured Dovecot's authentication mechanisms. This was done in `/etc/dovecot/conf.d/10-auth.conf`:



```
hemanthSNA@hemanthserver:~ — sudo nano /etc/dovecot/c...
GNU nano 5.6.1 /etc/dovecot/conf.d/10-auth.conf Modified
# plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
# gss-spnego
# NOTE: See also disable_plaintext_auth setting.
disable_plaintext_auth = no
auth_mechanisms = plain login

##
## Password and user databases
##

#
# Password database is used to verify user's password (and nothing more).
# You can have multiple passdbs and userdbs. This is useful if you want to
# allow both system users (/etc/passwd) and virtual users to login without
# duplicating the system users into virtual database.
#
# <doc/wiki/PasswordDatabase.txt>
#
# User database specifies where mails are located and what user/group IDs

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

Figure 19: `/etc/dovecot/conf.d/10-auth.conf`

disable_plaintext_auth = no: Allows plaintext authentication, which is safe to use when secure channels like SSL/TLS are enabled.

auth_mechanisms = plain login: Specifies the supported authentication mechanisms. The plain and login methods are widely supported by email clients and provide compatibility.

These settings ensure users can authenticate securely while still maintaining compatibility with most email clients.

3.2.3 Enabling SSL/TLS

To secure the communication between the email client and the server, I enabled SSL/TLS encryption. This was configured in `/etc/dovecot/conf.d/10-ssl.conf`

```
Complete!
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/dovecot.conf
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Figure 20: `sudo nano /etc/dovecot/conf.d/10-ssl.conf`

```
GNU nano 5.6.1 /etc/dovecot/conf.d/10-ssl.conf Modified
##
## SSL settings
##
# SSL/TLS support: yes, no, required. <doc/wiki/SSL.txt>
# disable plain pop3 and imap, allowed are only pop3+TLS, pop3s, imap+TLS and imaps
# plain imap and pop3 are still allowed for local connections
ssl = required

# PEM encoded X.509 SSL/TLS certificate and private key. They're opened before
# dropping root privileges, so keep the key file unreadable by anyone but
# root. Included doc/mkcert.sh can be used to easily generate self-signed
# certificate, just make sure to update the domains in dovecot-openssl.cnf
ssl_cert = </etc/ssl/certs/mailcert.pem
ssl_key = </etc/ssl/private/mailkey.pem

# If key file is password protected, give the password here. Alternatively
# give it when starting dovecot with -p parameter. Since this file is often
# world-readable, you may want to place this setting instead to a different
# root owned 0600 file by using ssl_key_password = <path>.
#ssl_key_password =

# PEM encoded trusted certificate authority. Set this only if you intend to use
# ssl_verify_client_cert=yes. The file should contain the CA certificate(s)
# followed by the matching CRL(s). (e.g. ssl_ca = </etc/pki/dovecot/certs/ca.pem>)
#ssl_ca =

# Require that CRL check succeeds for client certificates.
#ssl_require_crl = yes

# Directory and/or file for trusted SSL CA certificates. These are used only
# when Dovecot needs to act as an SSL client (e.g. imapc backend or
# submission service). The directory is usually /etc/pki/dovecot/certs in
# Debian-based systems and the file is /etc/pki/tls/cert.pem in
# RedHat-based systems. Note that ssl_client_ca_file isn't recommended with
# large CA bundles, because it leads to excessive memory usage.
#ssl_client_ca_dir =
#ssl_client_ca_file =

# Require valid cert when connecting to a remote server
#ssl_client_require_valid_cert = yes

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^J Execute    ^C Location   ^U Undo       ^M Set Mark   ^_ To Bracket
^X Exit      ^R Read File  ^\ Replace    ^V Paste      ^_ Justify    ^_ Go To Line ^E Redo       ^G Copy       ^_ Where Was
```

Figure 21: `/etc/dovecot/conf.d/10-ssl.conf`

ssl = required: Enables SSL/TLS for secure communication.

ssl_cert: Specifies the path to the SSL certificate file.

ssl_key: Specifies the path to the private key file.

I used the same self-signed SSL certificate and key that I generated earlier for Postfix. To ensure the certificate and key files were accessible by Dovecot, I checked their permissions:

```
[hemanthSNA@hemanthserver ~]$ sudo chmod 600 /etc/ssl/private/mailkey.pem
[hemanthSNA@hemanthserver ~]$ sudo chmod 644 /etc/ssl/certs/mailcert.pem
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Figure 22: enable permissions ssl

These permissions protect the private key while allowing Dovecot to read the certificate and key files.

3.2.4 Restarting Dovecot

After completing the configuration, I restarted the Dovecot service to apply the changes:

```
hemanthSNA@hemanthserver ~]$ sudo systemctl restart dovecot
[sudo] password for hemanthSNA:
[hemanthSNA@hemanthserver ~]$ sudo systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
[hemanthSNA@hemanthserver ~]$ sudo systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-01-23 12:47:19 EST; 24s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
  Main PID: 6619 (dovecot)
    Status: "v2.3.16 (7e2e900c1a) running"
     Tasks: 4 (limit: 10964)
    Memory: 5.1M
       CPU: 77ms
    CGroup: /system.slice/dovecot.service
            └─6619 /usr/sbin/dovecot -F
               └─6620 dovecot/anvil
                  └─6621 dovecot/log
                     └─6625 dovecot/config

Jan 23 12:47:19 hemanthserver.bungkus.net systemd[1]: Starting Dovecot IMAP/POP3 email server...
Jan 23 12:47:19 hemanthserver.bungkus.net dovecot[6619]: master: Dovecot v2.3.16 (7e2e900c1a) starting up for imap, pop3
Jan 23 12:47:19 hemanthserver.bungkus.net systemd[1]: Started Dovecot IMAP/POP3 email server.
[hemanthSNA@hemanthserver ~]$
```

Figure 23: Restart, enable and test Dovecot

I also ensured the service was enabled to start automatically on boot and tested using `sudo systemctl status dovecot` to ensure it is running.

3.2.5 Testing IMAP and POP3 Connectivity

To test Dovecot's functionality, I used `telnet` and `openssl` commands to connect to the IMAP and POP3 services:

```
[hemanthSNA@hemanthserver ~]$ openssl s_client -connect hemanthserver.bungkus.net:993
Connecting to 192.168.200.4
CONNECTED(00000003)
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify error:num=18:self-signed certificate
verify return:1
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify return:1
---
Certificate chain
 0 s:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
  i:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jan 23 17:26:32 2025 GMT; NotAfter: Jan 23 17:26:32 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIETITCCAZgwAwIBAgIUPI4J3X/Y5c13FyKneUb0zHxX+vsW0QYJKoZIhvcNAQEL
BQAwgbcUxCzA3BgNVBAYTAk1ZMREdWYDVQIDAhTZWxhbmdvcjEQAQA4GA1UEBmWw
UHVjaG9uZzEVMBMGA1UECgVMSGVtYW50aCBDb3JwMRYwFAYDVQQLDA1JVCBEZXh
cnRtZW50MSIwIAYDVQDDbGloZW1hbnRoc2VydmlvLnJlbnRdXmubmV0MS4wLAYj
KoZiIhvcNAQKBHh9hZG1pbkBoZW1hbnRoc2VydmlvLnJlbnRdXmubmV0MB4XD
T1I0QYMeZm3YzYzMDXDTI2MDEyMDEyZm3YzYzMDlwbGwbcUxCzA3BgNVBAYTAk1ZMREdWY
DVQIDAhTZWxhbmdvcjEQAQA4GA1UEBmWwUHVjaG9uZzEVMBMGA1UECgVMSGVtYW50
aCBDb3JwMRYwFAYDVQQLDA1JVCBEZXhcnRtZW50MSIwIAYDVQDDbGloZW1hbnRo
c2VydmlvLnJlbnRdXmubmV0MS4wLAYjKoZiIhvcNAQKBHh9hZG1pbkBoZW1hbnRo
c2VydmlvLnJlbnRdXmubmV0MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgK
CAQEAc2c46bhiP0KTjijG8f1WHBwZnmD4INuYQ4hH0Q99gnoJS/5I3IRK695SaRe
kEP0SUA4XgWhx5p4xM+CWRdmz9ekmB4Acced7IpaXC9qiaidtQKv2YRc322w70pe8n
gy/8Uku46X05Sax4c8qgPwNg5R70KTvEgY0vju1xmva0tzTZxW4+6RR+WkUP0/bP
7f50J3wTHHgBh3wxTG8npLG7Hb4THI9D0qPhsmVmjPtZSK58Wzjqllkb0J++a/DqL
oYyKzVEzyVDAHMA5KwptHpD9zu449Rag/3UN8gfw8tEXWQVAnHvVPLzopdWodttw
bPM0nF52BgIdQZFgko+Yv/Yd5QIDAQAB01MwUTAdBgNVHQ4EFgQUw/b+brbT0E8K
eW02o5aSwGnZkhdwHwYDVR0jBBGwFAUw/b+brbT0E8KeW02o5aSwGnZkhdwDwYD
VR0TAQH/BAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEACxwHrLAXP1KEB9Nm9wv
gsmYv+hhEtKfsYYOq7t7pz9gAnWKx8CZclvQu9fjLI+AvNGVYF1U0IisuhngshXZ
Kpqi00lv3ZRTcHrdIf+bp0lf0d4MctJ9crNm1++atKN2/L6C9LA7e+VXDOpbk1dtG
KpNMHxpyQy00Xe12CpnPnu+Ea2NTxCLewTniyoEN4Gis0Tca0hcDBR/MyPbI0n
8F6reM61wzK2bspc+vV/3ws3Cvm1fRGPQKwYzReQH9cD4/9ufqafW0E1r7TC/Ws
+s1Qr+CqEjYtFNFDh56kTMBzSyzHBYup/W7Xd7QZ5Dh0k+46aFk5tMHEuIk8aT
9A==
-----END CERTIFICATE-----
subject=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
issuer=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
---
```

Figure 24: `openssl s_client -connect hemanthserver.bungkus.net:993`

This command tests the IMAP service on port 993, verifying the SSL/TLS connection and server response.

```
[hemanthSNA@hemanthserver ~]$ openssl s_client -connect hemanthserver.bungkus.net:995
Connecting to 192.168.200.4
CONNECTED(00000003)
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify error:num=18:self-signed certificate
verify return:1
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify return:1
---
Certificate chain
 0 s:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
 1 i:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
 a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
 v:NotBefore: Jan 23 17:26:32 2025 GMT; NotAfter: Jan 23 17:26:32 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEITCCAzwGAwIBAgIUPI4J3X/Y5c13FyKneUb0zHxX+vsWQDYJKoZIhvcNAQEL
BQAwgBUXCzAJBgNVBAYTAk1ZMREwDwYDVQQIDAhTZWxhbmdvcjEOMAA4GA1UEBwwH
UHVjag9uZzEVMBMGA1UECgwwSGVtYW50aCBDb3JwMRVwFAYDVQQLEDA1JVCBEZXBh
cnRtZW50MSIwIAYDVQQDBloZW1hbnRoc2VydmlVYmJ1bmdrdXMubmV0MS4wLAYJ
KoZIhvcNAQkBFh9hZG1pbkBoZW1hbnRoc2VydmlVYmJ1bmdrdXMubmV0MBA4XDII
MDYyMzE3MjYzMDUxMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEy
VQ0IDAhtZWxhbmdvcjEOMAA4GA1UEBwwHUHVjag9uZzEVMBMGA1UECgwwSGVtYW50
aCBDb3JwMRVwFAYDVQQLEDA1JVCBEZXBhcnRtZW50MSIwIAYDVQQDBloZW1hbnRo
c2VydmlVYmJ1bmdrdXMubmV0MS4wLAYJKoZIhvcNAQkBFh9hZG1pbkBoZW1hbnRo
c2VydmlVYmJ1bmdrdXMubmV0MIIIBjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCA
QEAA2c46bhipOKTjijG8f1WHBw2nmD4INuYQ4hH0QN99gnoJS/5I3RIK695SaReK
ePOSu4XgWhx5p4xM+CWRdmz9ekmB4Acced7IpAx9qia1dtQKv2YRc322w70pe8n
gy/8Uku46X0Sax4c8qgPwNgSR70KTvEgY0vju1xmva0tzTZxXw4+6RR+WkUPo/bP
7s50JwTHHGBH3wxTG8nPLG7Hb4THI9DQpPhsmVMjPtZSK58Wbjqlkb0J++a/DqL
oYyKzVEzyVdaHMA5KWphtpD9zu449Rag/JUN8gfwBtEXWQVAnHvVPLzopdWodttw
bPM0nF52BG1dQZFKg0+Yv/Yd5QIDAQABo1MwUADBgNVHQ4EFgQUw/b+brbT0E8K
eW0Z5aSWGnZhkdwwHwYDVR0jBBGwFoAUw/b+brbT0E8Kew0Z5aSWGnZhkdwwDwYD
VR0TAQH/BAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEACxwhrLAXP1KEb9Nm9wv
gsmYv+hhEtKfsYVQq7t7p29gAnWkxC8ZclVgCu9fjLI+AvNGVYF1U0IsuhngshXz
KPq700lv3ZRTcHrdIf+b0l0fd4mJ9crNm1++atXN2/l6C0LxAt7e+VXDopbk1dtG
CKNVHxpYq90x0e12CpnPnu+Ea2NTXcZLewTn1yoEN4iGSoTca0hcDBR/MYpbI0n
8F6reM6lwzK2bspc+vV/3ws3Cvm1fRGPQwKyZReqH9cD4/9ufqafiW0E1R7TC/Ws
+slQ+CrEjYtNFNDh56kTMBzSyzHBYup/W7Xd7QZSDdh0K+46atFK5tMHEuWIK8At
9A==
-----END CERTIFICATE-----
subject=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
issuer=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
---
```

Figure 25: openssl s_client -connect hemanthserver.bungkus.net:995

Similarly, this command tests the POP3 service on port 995.

Installing and configuring Dovecot was a critical step in enabling users to access their emails securely and efficiently. By carefully editing the modular configuration files, I ensured that Dovecot could handle IMAP and POP3 protocols, authenticate users, and integrate seamlessly with Postfix. This process not only deepened my understanding of mail server components but also highlighted the importance of secure communication and precise configuration in email systems (jproactive, 2023).

4.0 Configuration of SSL/TLS Encryption

Before we go into the configuration, it's crucial to understand why SSL/TLS is required. In the absence of encryption, data exchanged between email clients and the server is exposed to interception and spotlight by criminal entities (Gomez, 2023). Enabling SSL/TLS encrypts all communication, safeguarding the confidentiality and integrity of the data. I utilized a self-signed certificate for my configuration, which is appropriate for testing and internal environments.

4.1 Generating SSL/TLS Certificates

To enable SSL/TLS encryption, I first needed to create a self-signed SSL certificate and a private key. I generated these using the OpenSSL tool with the following command:

[illegible]

Figure 26: `sudo openssl req -new -x509 -days 365 -nodes -out /etc/ssl/certs/mailcert.pem -keyout /etc/ssl/private/mailkey.pem`

Here's what each part of the command does:

- **openssl:** Invokes the OpenSSL tool for managing SSL certificates.
- **req -new -x509:** Generates a new certificate signing request and creates a self-signed certificate.
- **-days 365:** Specifies that the certificate will be valid for 365 days.
- **-nodes:** Ensures the private key is not encrypted with a passphrase, which simplifies server operations.
- **-out:** Specifies the output path for the certificate file (mailcert.pem).

- **-keyout:** Specifies the output path for the private key file (mailkey.pem).

During the process, I was prompted to provide several details for the certificate:

- **Country Name (C):** Entered my country code (e.g., MY for Malaysia).
- **State or Province Name (ST):** Specified my state (Selangor).
- **Locality Name (L):** Provided my city name (Puchong).
- **Organization Name (O):** Entered the name of my organization or company (Hemanth Corp).
- **Organizational Unit Name:** This is optional but in my case I provided something related to my course (IT Department).
- **Common Name (CN):** This was the most important field, where I entered my server's FQDN (hemanthserver.bungkus.net).

4.2 Setting Permissions for SSL/TLS Files

Setting Permissions for SSL/TLS Files

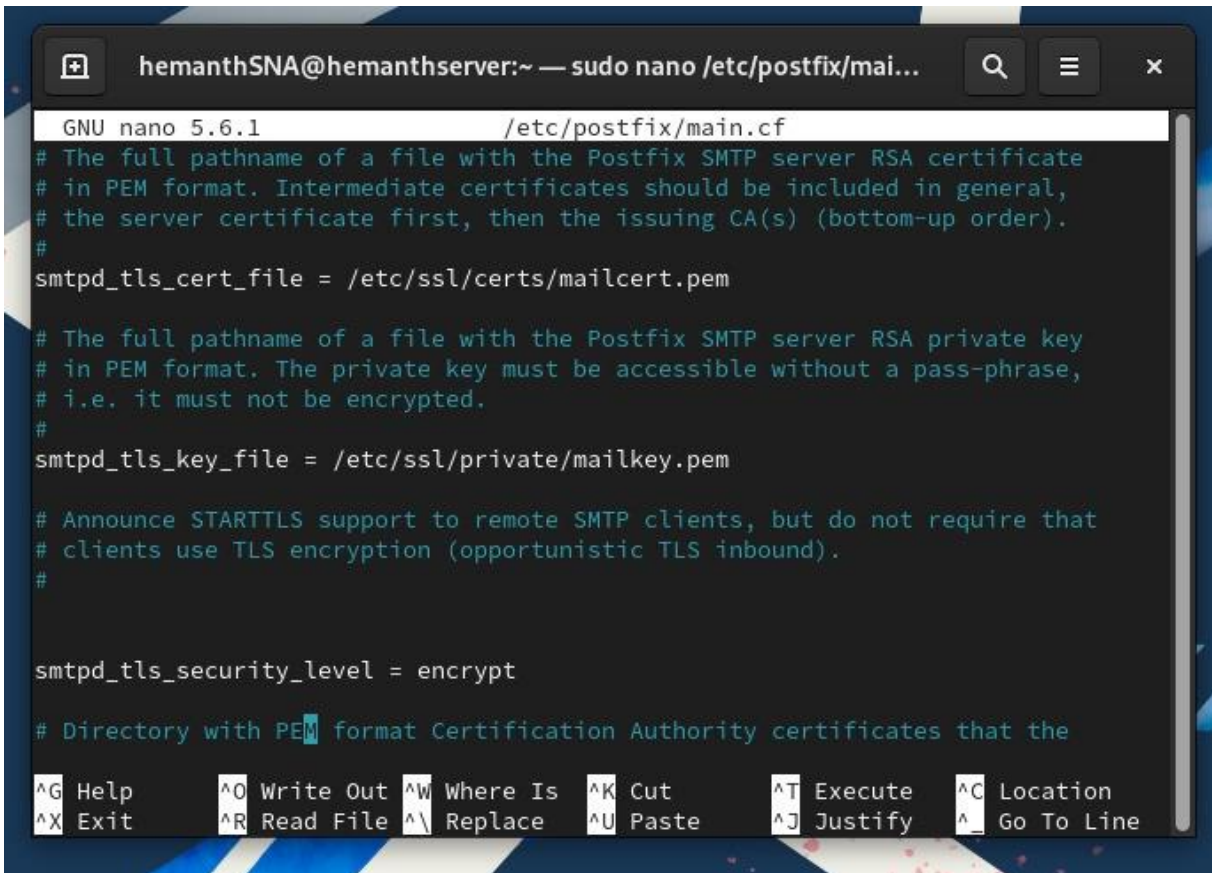
```
[hemanthSNA@hemanthserver ~]$ sudo chmod 600 /etc/ssl/private/mailkey.pem
[hemanthSNA@hemanthserver ~]$ sudo chmod 644 /etc/ssl/certs/mailcert.pem
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Figure 26: Sudo chmod 600 and 644

`sudo chmod 600 /etc/ssl/private/mailkey.pem` command ensures that only the root user can access the private key and the other one, `sudo chmod 644 /etc/ssl/certs/mailcert.pem` allows the certificate to be read by other processes, such as Postfix and Dovecot, while still maintaining security.

4.3 Configuring SSL/TLS in Postfix

Next, I updated the Postfix configuration file (/etc/postfix/main.cf) to enable SSL/TLS encryption (previously mentioned in 2.2.6 Enabling Secure Connections):



```
hemanthSNA@hemanthserver:~ — sudo nano /etc/postfix/mai...
GNU nano 5.6.1 /etc/postfix/main.cf
# The full pathname of a file with the Postfix SMTP server RSA certificate
# in PEM format. Intermediate certificates should be included in general,
# the server certificate first, then the issuing CA(s) (bottom-up order).
#
smtpd_tls_cert_file = /etc/ssl/certs/mailcert.pem

# The full pathname of a file with the Postfix SMTP server RSA private key
# in PEM format. The private key must be accessible without a pass-phrase,
# i.e. it must not be encrypted.
#
smtpd_tls_key_file = /etc/ssl/private/mailkey.pem

# Announce STARTTLS support to remote SMTP clients, but do not require that
# clients use TLS encryption (opportunistic TLS inbound).
#

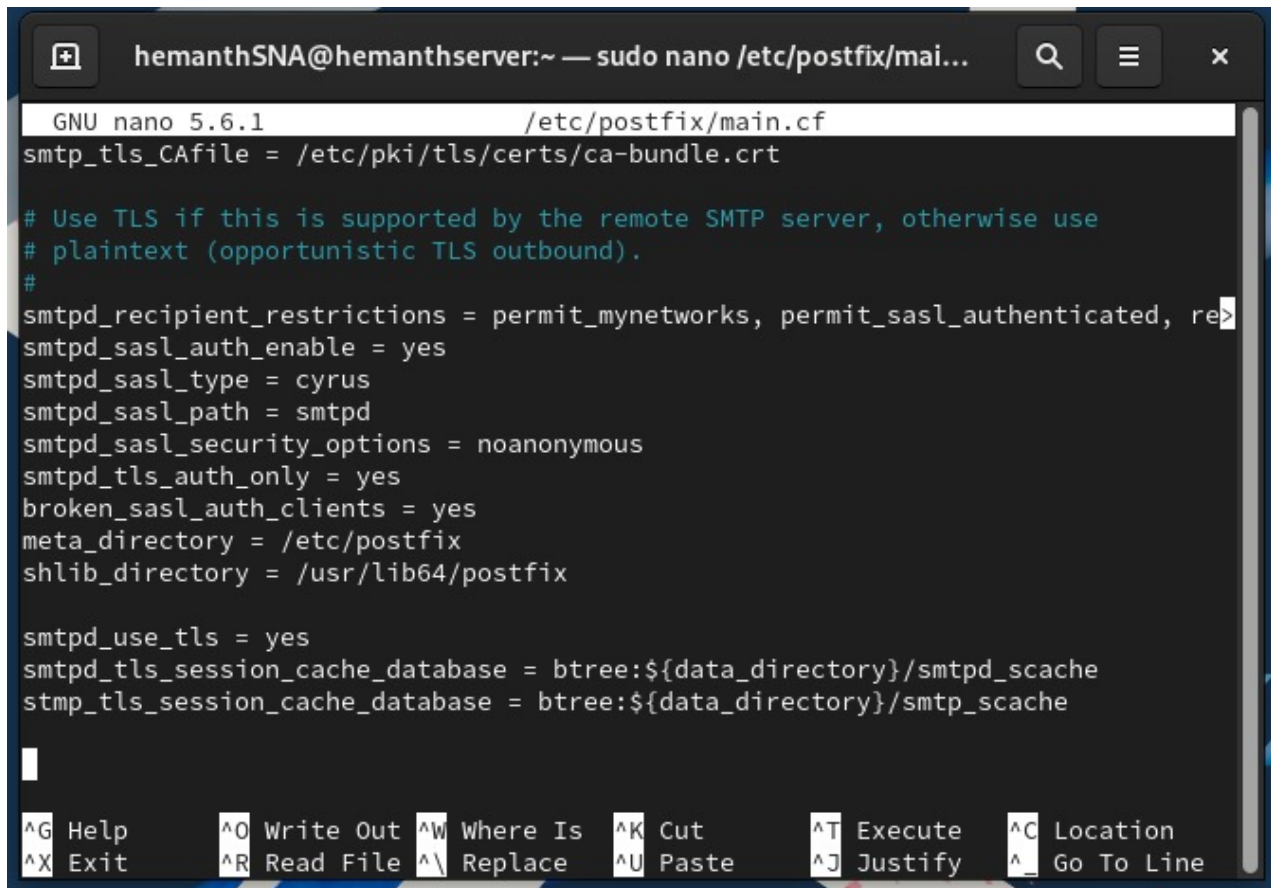
smtpd_tls_security_level = encrypt

# Directory with PEM format Certification Authority certificates that the

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Figure 27: sudo nano /etc/postfix/main.cf

- **smtpd_tls_cert_file**: Specifies the path to the SSL certificate file.
- **smtpd_tls_key_file**: Specifies the path to the private key file.
- **smtpd_tls_security_level = encrypt**: Forces encryption for all SMTP connections, ensuring secure transmission.



```
GNU nano 5.6.1 /etc/postfix/main.cf
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt

# Use TLS if this is supported by the remote SMTP server, otherwise use
# plaintext (opportunistic TLS outbound).
#
smtpd_recipient_restrictions = permit_mynetworks, permit_sasl_authenticated, re>
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = cyrus
smtpd_sasl_path = smtpd
smtpd_sasl_security_options = noanonymous
smtpd_tls_auth_only = yes
broken_sasl_auth_clients = yes
meta_directory = /etc/postfix
shlib_directory = /usr/lib64/postfix

smtpd_use_tls = yes
smtpd_tls_session_cache_database = btree:${data_directory}/smtpd_scache
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

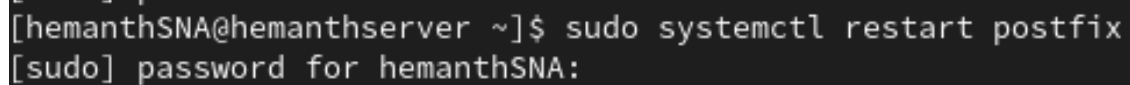
Figure 28: sudo nano /etc/postfix/main.cf (cont.)

- **smtpd_recipient_restrictions = permit_mynetworks, permit_sasl_authenticated, reject_unauth_destination:** Defines the rules for determining whether to accept or reject mail for a particular recipient.
- **smtpd_sasl_auth_enable = yes:** Enables SASL authentication for incoming mail.
- **smtpd_sasl_type = cyrus:** Specifies that Cyrus SASL is used for authentication.
- **smtpd_sasl_path = smtpd:** Specifies the path to the SASL library.
- **smtpd_sasl_security_options = noanonymous:** Disables anonymous authentication.
- **smtpd_tls_auth_only = yes:** Requires that TLS encryption is used for all authenticated connections.
- **broken_sasl_auth_clients = yes:** Enables support for clients that may not implement SASL correctly.

- **meta_directory** = **/etc/postfix**: Specifies the directory where Postfix stores its configuration files.
- **shlib_directory** = **/usr/lib64/postfix**: Specifies the directory where Postfix's shared libraries are located.
- **smtpd_use_tls** = **yes**: Enables TLS for SMTP communication.
- **smtpd_tls_session_cache_database** = **btree:\${data_directory}/smtpd_scache**: Configures a session cache specifically for TLS sessions when Postfix is acting as an SMTP server (i.e., receiving incoming mail).
- **smtp_tls_session_cache_database** = **btree:\${data_directory}/smtp_scache**: Configures a session cache for TLS sessions when Postfix is acting as an SMTP client (i.e., sending outgoing mail).

These settings ensure that Postfix uses SSL/TLS for both incoming and outgoing mail, protecting user credentials and email data.

After editing the configuration, I restarted Postfix to apply the changes:



```
[hemanthSNA@hemanthserver ~]$ sudo systemctl restart postfix
[sudo] password for hemanthSNA:
```

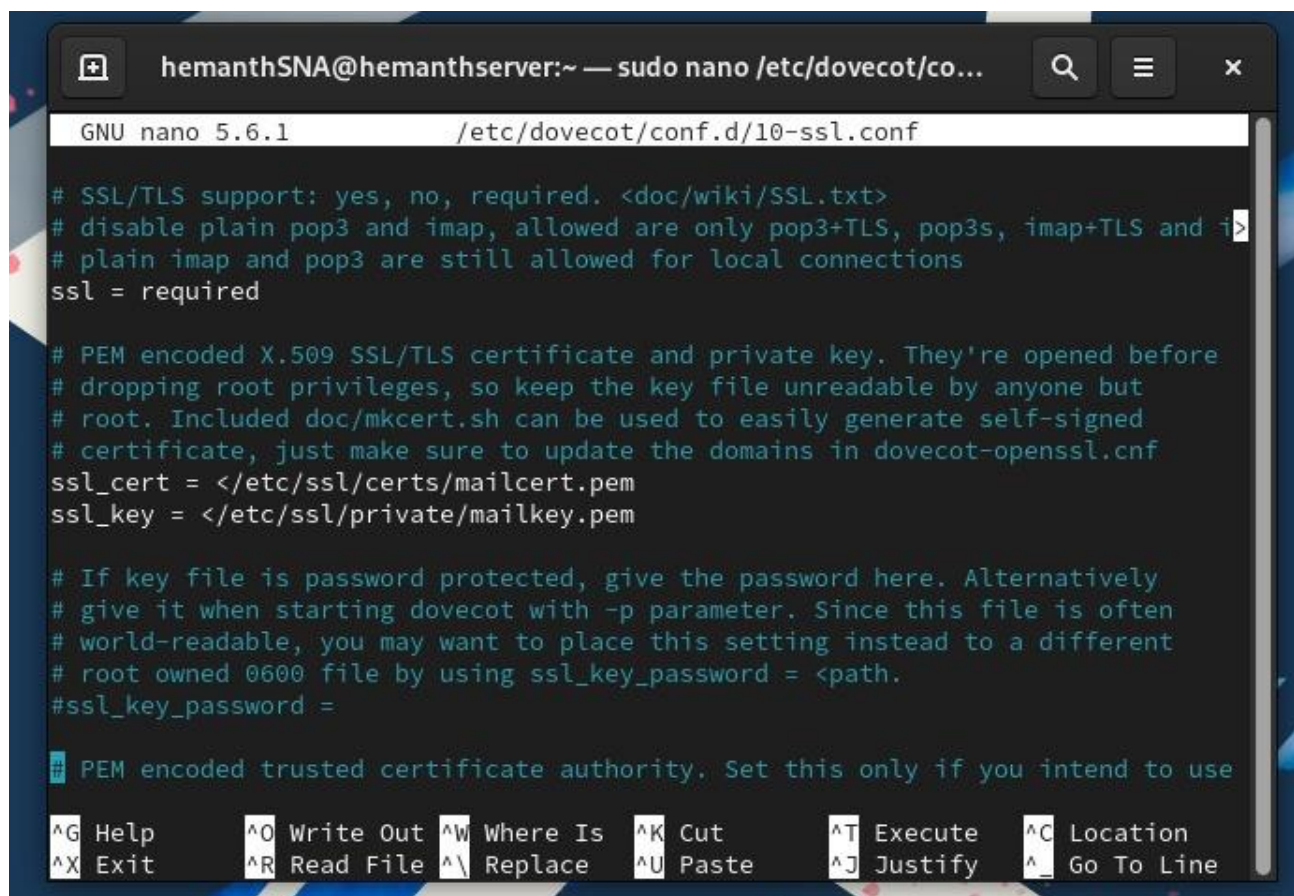
Figure 29: sudo systemctl restart postfix

4.4 Configuring SSL/TLS in Dovecot

To enable SSL/TLS for IMAP and POP3 in Dovecot, I updated the SSL configuration in `/etc/dovecot/conf.d/10-ssl.conf` (Previously mentioned in 3.2.3 Enabling SSL/TLS):

```
Complete!  
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/dovecot.conf  
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

Figure 30: Figure 20: `sudo nano /etc/dovecot/conf.d/10-ssl.conf`



The screenshot shows a terminal window titled "hemanthSNA@hemanthserver:~ — sudo nano /etc/dovecot/co...". The editor is GNU nano 5.6.1 editing the file /etc/dovecot/conf.d/10-ssl.conf. The content of the file is as follows:

```
# SSL/TLS support: yes, no, required. <doc/wiki/SSL.txt>  
# disable plain pop3 and imap, allowed are only pop3+TLS, pop3s, imap+TLS and i  
# plain imap and pop3 are still allowed for local connections  
ssl = required  
  
# PEM encoded X.509 SSL/TLS certificate and private key. They're opened before  
# dropping root privileges, so keep the key file unreadable by anyone but  
# root. Included doc/mkcert.sh can be used to easily generate self-signed  
# certificate, just make sure to update the domains in dovecot-openssl.cnf  
ssl_cert = </etc/ssl/certs/mailcert.pem  
ssl_key = </etc/ssl/private/mailkey.pem  
  
# If key file is password protected, give the password here. Alternatively  
# give it when starting dovecot with -p parameter. Since this file is often  
# world-readable, you may want to place this setting instead to a different  
# root owned 0600 file by using ssl_key_password = <path>.  
#ssl_key_password =  
  
# PEM encoded trusted certificate authority. Set this only if you intend to use
```

At the bottom of the terminal, there is a row of keyboard shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, ^_ Go To Line.

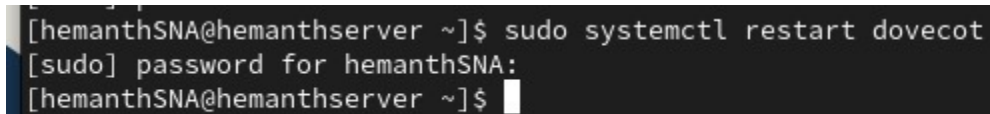
Figure 31: `/etc/dovecot/conf.d/10-ssl.conf`

ssl = yes: Enables SSL/TLS encryption for IMAP and POP3.

ssl_cert: Specifies the path to the SSL certificate file.

ssl_key: Specifies the path to the private key file.

After updating the configuration, I restarted Dovecot:



```
[hemanthSNA@hemanthserver ~]$ sudo systemctl restart dovecot
[sudo] password for hemanthSNA:
[hemanthSNA@hemanthserver ~]$
```

Figure 32: sudo systemctl restart dovecot

4.5 Testing SSL/TLS Configuration

To verify that SSL/TLS was working correctly, I used the `openssl s_client` command to test the secure connections for SMTP, IMAP, and POP3 (previously mentioned in Testing IMAP and POP3 Connectivity).

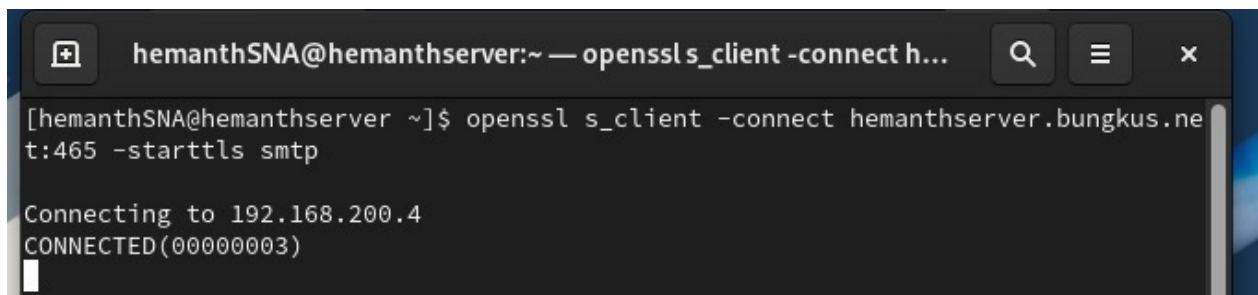
A terminal window titled 'hemanthSNA@hemanthserver:~ — openssl s_client -connect h...' shows the command `openssl s_client -connect hemanthserver.bungkus.net:465 -starttls smtp` being executed. The output indicates a successful connection to IP 192.168.200.4 on port 465, showing 'CONNECTED(00000003)'.

Figure 33: Testing SMTP (Port 465)

`openssl s_client -connect hemanthserver.bungkus.net:465 -starttls smtp`: Checks if SMTP connections are encrypted using SSL/TLS. A successful handshake indicates the configuration is correct.

```
[hemanthSNA@hemanthserver ~]$ openssl s_client -connect hemanthserver.bungkus.net:993
Connecting to 192.168.200.4
CONNECTED(00000003)
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify error:num=18:self-signed certificate
verify return:1
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify return:1
---
Certificate chain
 0 s:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
  i:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
  a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
  v:NotBefore: Jan 23 17:26:32 2025 GMT; NotAfter: Jan 23 17:26:32 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIITETCCAZwAwIBAgIUPIA3J3X/Y5c13FyKneUboZHXx+vsWdQYJKoZIhvcNAQEL
BQAwgBuXCAJBgNVBAYTAk1ZMREwDwYDVQQIDAhTZWxhbmdvcjEQMA4GA1UEBwwH
UHVjaG9uZzEVMGMGA1UECgwMSGVtYW50aC80b3JwMRwwFAYDVQQDLDA1JVCBEZXBh
cnRtZW50MSIwIAYDVQQDD0B1oZW1hbnRoc2VydmlvYmJ1bmdrdXMubmV0MS4wLmVJ
KoZIhvcNAQkBFh9hZG1pbkBoZW1hbnRoc2VydmlvYmJ1bmdrdXMubmV0MB4XDTE1
MDEyMzE3MjYzLW1oXDTI1MDEyMzE3MjYzLW1oZG1pbkBoZW1hbnRoc2VydmlvYmJ1bmdrdXMubmV0MS4wLmVJ
VQIDAHTZwxbmdvcjEQMA4GA1UEBwwHUHVjaG9uZzEVMGMGA1UECgwMSGVtYW50
aC80b3JwMRwwFAYDVQQDLDA1JVCBEZXBhcnRtZW50MSIwIAYDVQQDD0B1oZW1hbnRo
c2VydmlvYmJ1bmdrdXMubmV0MS4wLmVJ
KoZIhvcNAQkBFh9hZG1pbkBoZW1hbnRoc2VydmlvYmJ1bmdrdXMubmV0MIIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCGKC
AQEA2c46bh1pOKTjijG8f1WHBw2nmD4INuYQ4hH0QN99gnoJS/5I3RIK6955SaReK
eP0Su4Xgwhx5p4xM+CWRdmz9ekmB4Acced7IpaXc9qiaidtQKv2YRc322w70pe8n
gY/8Uku46X0Sax4c8qgPwNgSR70KTvEgY0vju1xmva0tzTZxXw4+6RR+WkUPO/bP
7s50JwTHHGbH3wxTg8npL67Hb4THI9DQpPhsmVMjPtZSK58W8jqlkboJ++a/DqL
oYyKzVEzyVDAHMA5KwtpD9zu449Rag/JUN8gfWbTExWQVAnHvVP1zopdwodttw
bPM0nF52BG1dQZFkg0+Yv/Yd5QIDAQABo1MwUTAdBgNVHQ4EFgQUw/b+brbT0E8K
eW0z5aSWGnZhkdwHwYDVR0jBBgwFoAUw/b+brbT0E8KeW0z5aSWGnZhkdwHwYD
VR0TAQH/BAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEACxwLXAP1KEB9Nm9wdu
gsmv+hhEtKfsYYOq7T7pZ9gAnWkXC8ZclVqCu9fjLI+AvNGVYF1U0IsuhngshXz
KPq00lv3ZRTcHrdIf+boL0fd4mJT9crNm1++atXN2/16C0LxA7e+VXDopbk1dtG
CkNVHxpyQy00xQe12CpnPnu+Ea2NTxCzLewTniyoEN4IGSoTca0hcDBR/MypbIOn
8F6reM61wzK2bspc+vV/3ws3Cvm1fRGpQwKyZReqH9cD4/9ufqafiW0EiR7TC/Ws
+slQ+CrEjYtNFDH56kTMBzSyzHBYup/w7Xd7QZSDdh0K+46atFK5tMHEuWIk8aT
9A==
-----END CERTIFICATE-----
subject=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
issuer=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
---
```

Figure 34: Testing IMAP (Port 993)

This tests the IMAP service over SSL/TLS, ensuring emails can be securely retrieved.

```
[hemanthSNA@hemanthserver ~]$ openssl s_client -connect hemanthserver.bungkus.net:995
Connecting to 192.168.200.4
CONNECTED(00000003)
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify error:num=18:self-signed certificate
verify return:1
depth=0 C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
verify return:1
---
Certificate chain
 0 s:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
 1 i:C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
 a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
 v:NotBefore: Jan 23 17:26:32 2025 GMT; NotAfter: Jan 23 17:26:32 2026 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIEITCCAzwGAwIBAgIUPI4J3X/Y5c13FyKneUb0zHxx+vsWQYJkOZiHvcNAQEL
BQAwgBuxCzAJBgNVBAYTAk1ZMRwEwYwDVQQIDAhTZWxhbmdvcjEQMA4GA1UEBwwH
UHVjaG9uZzEVMGMGA1UECgwMSGVtYW50aCBDb3JwMRwwFAYDVQQLEA1JVCBEZXhh
cnRtZW50MSIwIAAYDQYDQDBloZW1hbnRoc2VydmlVYmJ1bmdrdXMubmV0MS4wLWYJ
KoZiHvcNAQkBFh9hZG1pbkBoZW1hbnRoc2VydmlVYmJ1bmdrdXMubmV0MB4XDTI1
MDYyMzE3MjYzMDUxMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEyMDEy
VQYIDAhTZWxhbmdvcjEQMA4GA1UEBwwHUUHVjaG9uZzEVMGMGA1UECgwMSGVtYW50
aCBDb3JwMRwwFAYDVQQLEA1JVCBEZXhhcnRtZW50MSIwIAAYDQYDQDBloZW1hbnRo
c2VydmlVYmJ1bmdrdXMubmV0MS4wLWYJKoZiHvcNAQkBFh9hZG1pbkBoZW1hbnRo
c2VydmlVYmJ1bmdrdXMubmV0MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKC
AQEA2c46bh1pOKTjijG8f1WHBw2nmD4INuYQ4H0Q99gnoJS/5I3RIK695SaReK
eP0Su4XgWxx5p4xM+CWRdmz9ekmB4Acced7IPaXC9qia1dtQKv2YRc322w70pe8n
gy/8Uku46X0Sax4c8qgPwNgSR70KTvEgY0vju1xmva0tzTZxXw4+6RR+WkUPo/bP
7s503wTHHGBH3wxT68npLG7Hb4THI9DQpPhsmVMjPtZSK58W8jqlkb0J++a/DqL
oYyKzVEzyVdaHMA5KwptpD9zu449Rag/JUN8gfwBtEXWQVAnHvVPLzopdWodttw
bPM0nF52BG1dQZFKg0+Yv/Yd5QIDAQABO1MwUADBgNVHQ4EFgQUw/b+brbT0E8K
eW0Z5aSWGnZhkdwwHwYDVR0jBBGwFoAUw/b+brbT0E8KeW0Z5aSWGnZhkdwwDwYD
VR0TAQH/BAUwAwEB/zANBgkqhkiG9w0BAQsFAAOCAQEAxwLAXP1KEb9Nm9wvdx
gsmYv+hhEtKfsYVQq7t7p29gAnWkxC8ZclVgCu9fjLI+AvNGVYF1U0IsuhngshXz
KPq700lv3ZRTcHrdIf+bo10fd4mJ79crNm1++atXN2/l6C0LxAt7e+VXDopbk1dtG
CKNVHxpyQy00x0e12CpnPnu+Ea2NTxCzLewTn1yoEN41GSoTca0hcDBR/MYpbI0n
8F6reM61wzK2bspc+vV/3ws3Cvm1fRGpQwKyZReqH9cD4/9ufqaf1W0E1R7TC/Ws
+slQ+CrEjYtNFNDh56kTMBzSyzHBYup/W7Xd7QZ5Ddh0K+46atFK5tMHEuWIk8aT
9A==
-----END CERTIFICATE-----
subject=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
issuer=C=MY, ST=Selangor, L=Puchong, O=Hemanth Corp, OU=IT Department, CN=hemanthserver.bungkus.net, emailAddress=admin@hemanthserver.bungkus.net
---
```

Figure 35: Testing POP3 (Port 995)

This verifies the POP3 service is secured by SSL/TLS.

For each test, I checked the server's certificate details and verified that the connection was encrypted.

4.6 Configuring Thunderbird to Use SSL/TLS

To complete the setup, I configured Thunderbird on my client machine to use SSL/TLS for both IMAP and SMTP:

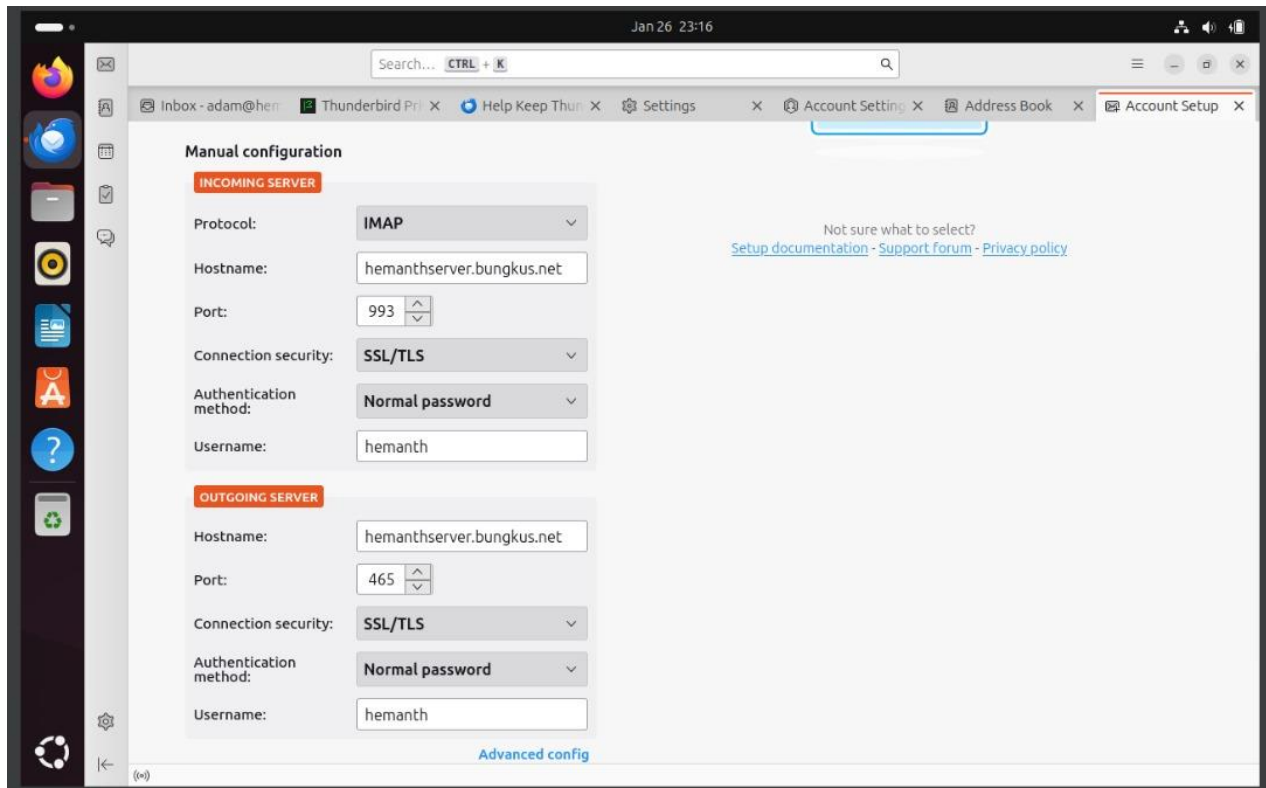


Figure 36: Manual Configuration Thunderbird

To complete the setup, I configured Thunderbird on my client machine to use SSL/TLS for both IMAP and SMTP:

1. Incoming Server (IMAP):

1.1. Server Name: hemanthserver.bungkus.net

1.2. Port: 993

- 1.3. Connection Security: SSL/TLS
- 1.4. Authentication: Normal Password
- 2. **Outgoing Server (SMTP):**
 - 2.1. Server Name: hemanthserver.bungkus.net
 - 2.2. Port: 465
 - 2.3. Connection Security: SSL/TLS
 - 2.4. Authentication: Normal Password

Thunderbird prompted me to trust the self-signed certificate. I manually accepted the certificate to establish a secure connection.

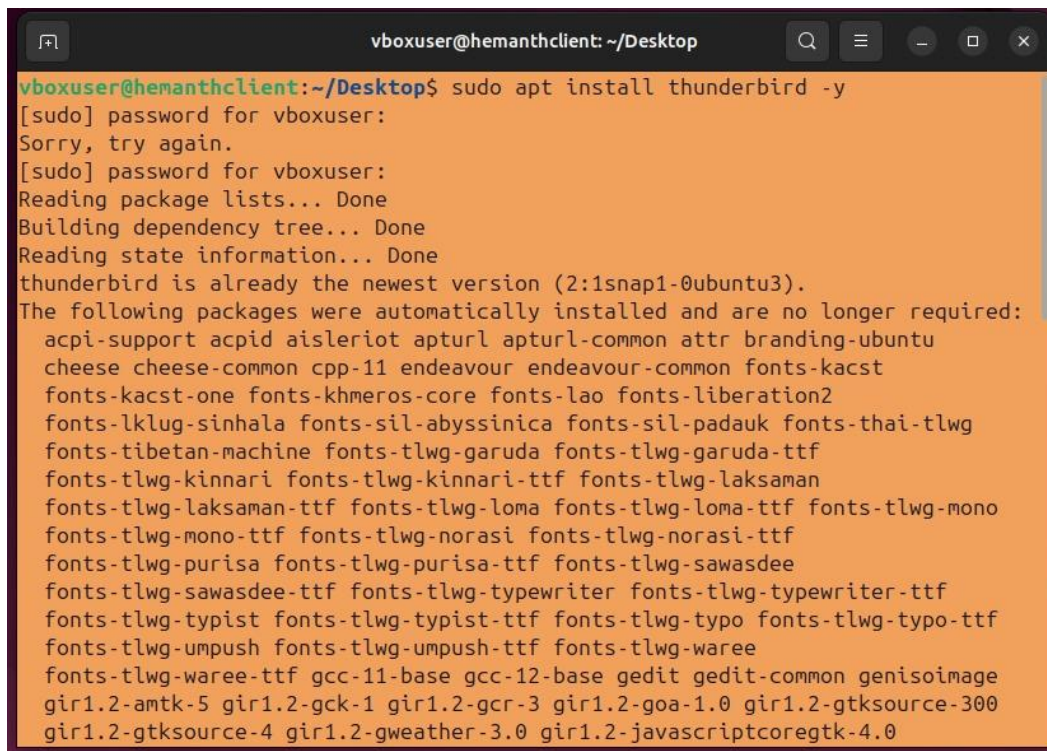
Configuring SSL/TLS encryption was a crucial step in securing my email server. By generating a self-signed certificate and integrating it into Postfix and Dovecot, I ensured that all communication between the server and email clients was encrypted. Testing the setup with tools like openssl_s_client provided confidence in the security of the configuration. This process highlighted the importance of encryption in protecting sensitive information and maintaining user trust in email systems.

5.0 Troubleshooting and Testing Email Functionality Using Thunderbird

After configuring Postfix, Dovecot, and SSL/TLS encryption, the next critical step was to test the functionality of my email server. I used Thunderbird, a widely-used third-party email client, to verify that the server could send and receive emails securely. This process involved configuring Thunderbird, testing email functionality, and troubleshooting any issues that arose during the process. Below, I detail each step I performed, along with explanations and observations.

5.1 Installing Thunderbird

To begin testing, I installed Thunderbird on my client machine, which was running Ubuntu. I used the following command to install it via the package manager:

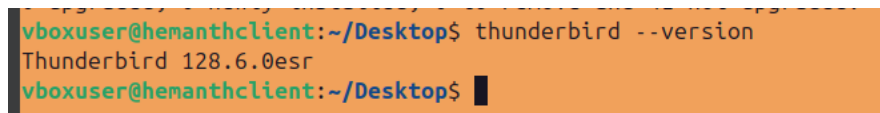


```
vboxuser@hemanthclient: ~/Desktop
vboxuser@hemanthclient:~/Desktop$ sudo apt install thunderbird -y
[sudo] password for vboxuser:
Sorry, try again.
[sudo] password for vboxuser:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
thunderbird is already the newest version (2:1snap1-0ubuntu3).
The following packages were automatically installed and are no longer required:
  acpi-support acpid aisleriot apturl apturl-common attr branding-ubuntu
  cheese cheese-common cpp-11 endeavour endeavour-common fonts-kacst
  fonts-kacst-one fonts-khmeros-core fonts-lao fonts-liberation2
  fonts-lklug-sinhala fonts-sil-abyssinica fonts-sil-padauk fonts-thai-tlwg
  fonts-tibetan-machine fonts-tlwg-garuda fonts-tlwg-garuda-ttf
  fonts-tlwg-kinnari fonts-tlwg-kinnari-ttf fonts-tlwg-laksaman
  fonts-tlwg-laksaman-ttf fonts-tlwg-loma fonts-tlwg-loma-ttf fonts-tlwg-mono
  fonts-tlwg-mono-ttf fonts-tlwg-norasi fonts-tlwg-norasi-ttf
  fonts-tlwg-purisa fonts-tlwg-purisa-ttf fonts-tlwg-sawasdee
  fonts-tlwg-sawasdee-ttf fonts-tlwg-typewriter fonts-tlwg-typewriter-ttf
  fonts-tlwg-typist fonts-tlwg-typist-ttf fonts-tlwg-typo fonts-tlwg-typo-ttf
  fonts-tlwg-umpush fonts-tlwg-umpush-ttf fonts-tlwg-waree
  fonts-tlwg-waree-ttf gcc-11-base gcc-12-base gedit gedit-common genisoimage
  gir1.2-atk-5 gir1.2-gck-1 gir1.2-gcr-3 gir1.2-goa-1.0 gir1.2-gtksource-300
  gir1.2-gtksource-4 gir1.2-gweather-3.0 gir1.2-javascriptcoregtk-4.0
```

Figure 37: Installing Thunderbird

- **sudo:** Ensures I have administrative privileges for the installation.
- **apt install thunderbird:** Fetches and installs Thunderbird along with its dependencies.
- **-y:** Automatically confirms the installation process, avoiding manual prompts.

This installed Thunderbird and added it to the system's application menu, allowing me to launch it directly.



```
vboxuser@hemanthclient:~/Desktop$ thunderbird --version
Thunderbird 128.6.0esr
vboxuser@hemanthclient:~/Desktop$
```

Figure 38: Thunderbird version

I also checked if my Thunderbird was running on the latest software updates and patches to minimize the risks of running into bugs and errors. My Thunderbird version is currently at the

latest patch as of January 27th 2025 and can be verified on the official Thunderbird website which shows the version to be 128.6.0esr that was released on January 8th 2025. (Thunderbird, 2025).

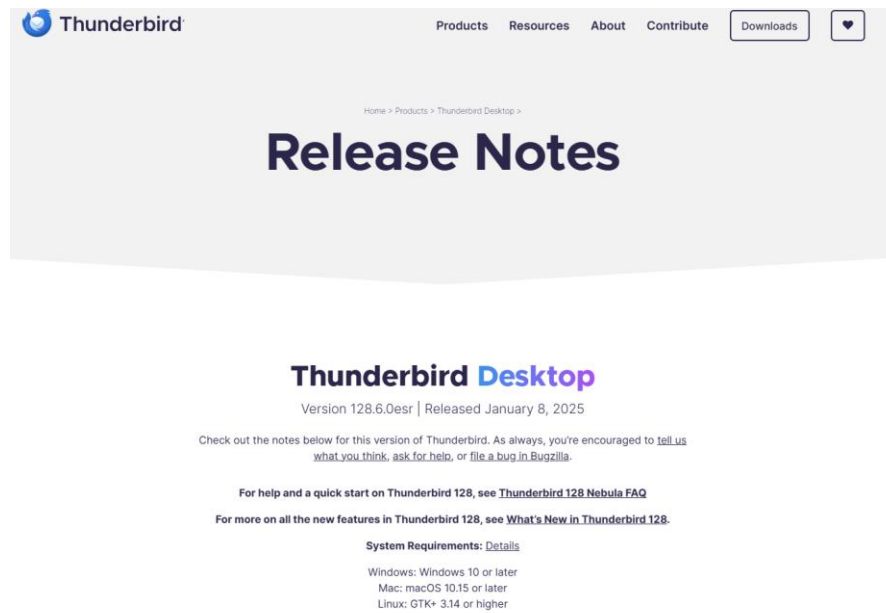


Figure 39: Latest Thunderbird version

5.2 Configuring Thunderbird

Once Thunderbird was installed, I launched it and set up my email account to connect to the server. During the setup, I was prompted to enter the following information:

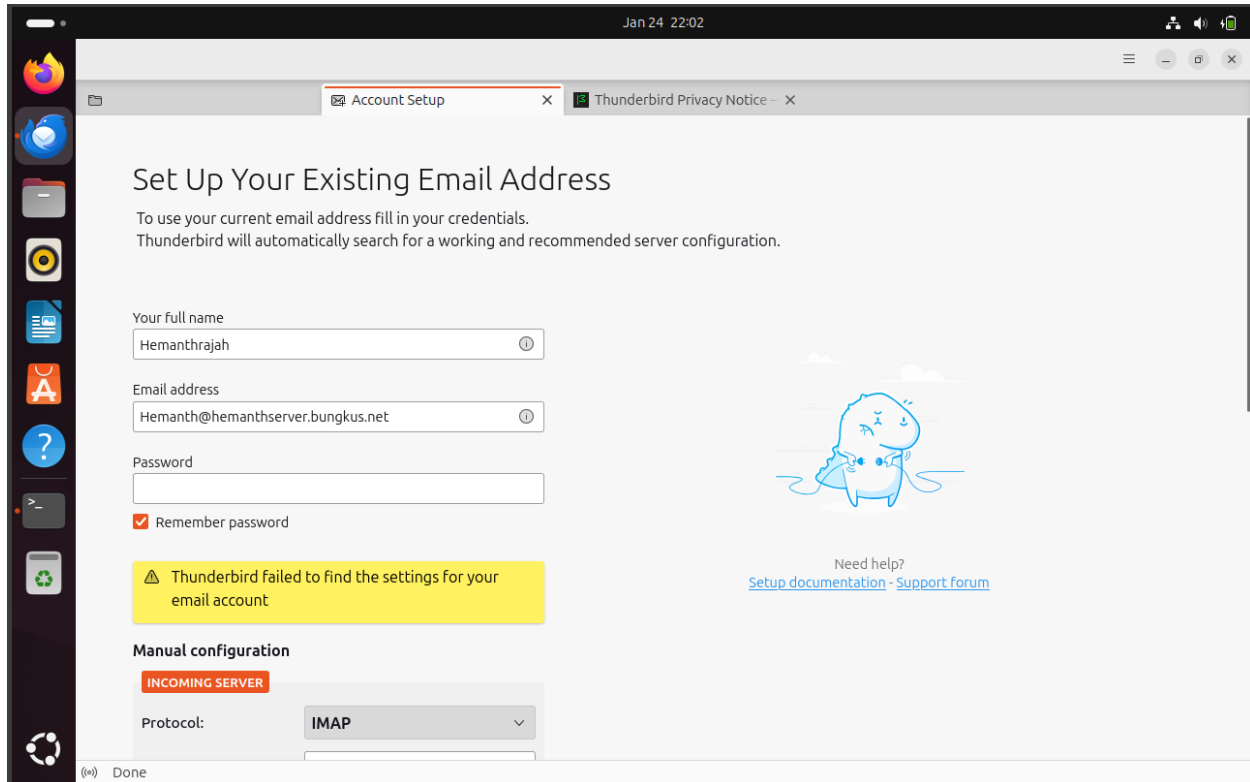


Figure 40: Set Up Thunderbird

- **Your Name:** I entered my display name (e.g., Hemanthrajah).
- **Email Address:** I provided my full email address (e.g., hemanth@bungkus.net).
- **Password:** I entered the password for the email account I had previously created on the server.

Thunderbird attempted to automatically detect the server settings, but I chose to configure them manually to ensure accuracy.

5.2.1 Manually Configuring Server Settings

During manual configuration, I specified the incoming and outgoing server details to match the settings of my email server (previously mentioned in 4.6 Configuring Thunderbird to Use SSL/TLS):

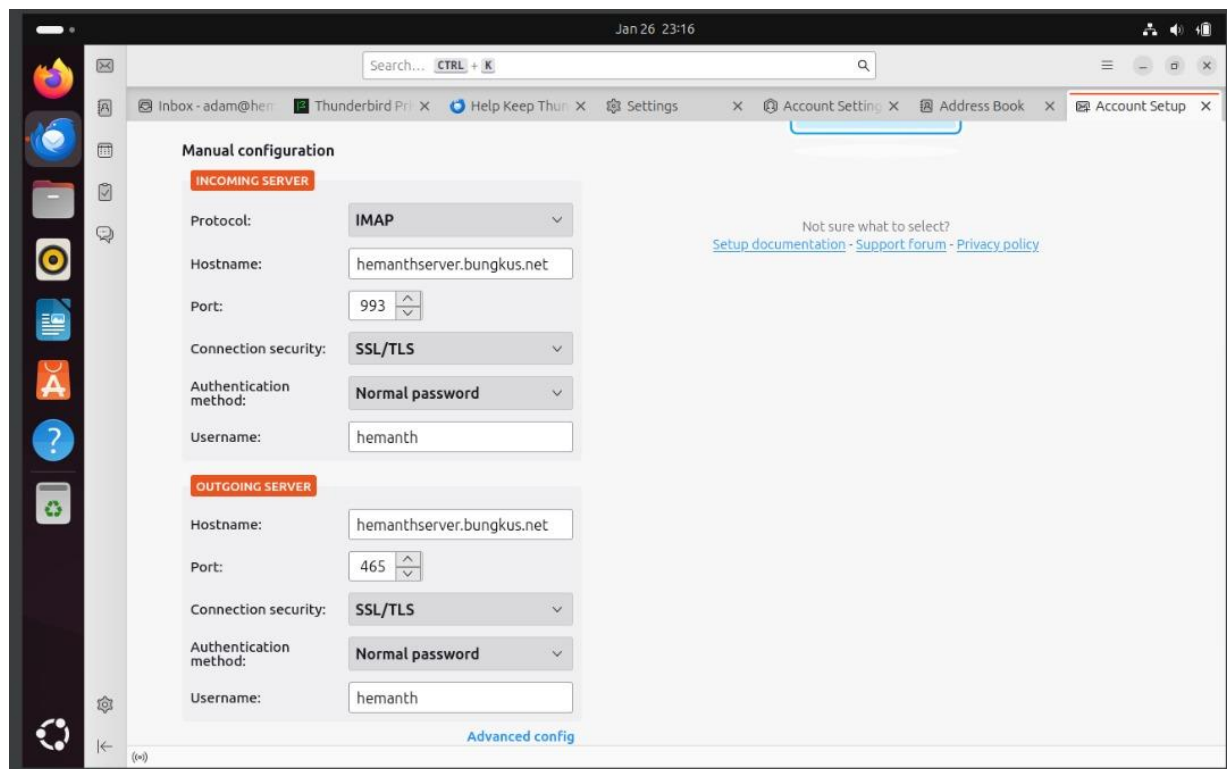


Figure 41: Manual Configuration of Thunderbird

1. Incoming Server (IMAP):

- 1.1. Server Name: hemanthserver.bungkus.net
- 1.2. Port: 993 (for IMAP over SSL/TLS)
- 1.3. Connection Security: **SSL/TLS**
- 1.4. Authentication Method: **Normal Password**
- 1.5. Username: : The user account created in Rockey server (e.g., hemanth).

2. Outgoing Server (SMTP):

- 2.1. Server Name: hemanthserver.bungkus.net
- 2.2. Port: 465 (for SMTP over SSL/TLS)
- 2.3. Connection Security: **SSL/TLS**
- 2.4. Authentication Method: **Normal Password**
- 2.5. Username: The user account created in Rockey server (e.g., hemanth).

These settings ensure that all communication between Thunderbird and the email server is encrypted and uses secure authentication.

5.2.2 Testing Email Sending Functionality

To test the outgoing mail functionality, I composed a test email in Thunderbird and sent it to another email account. I observed the following:

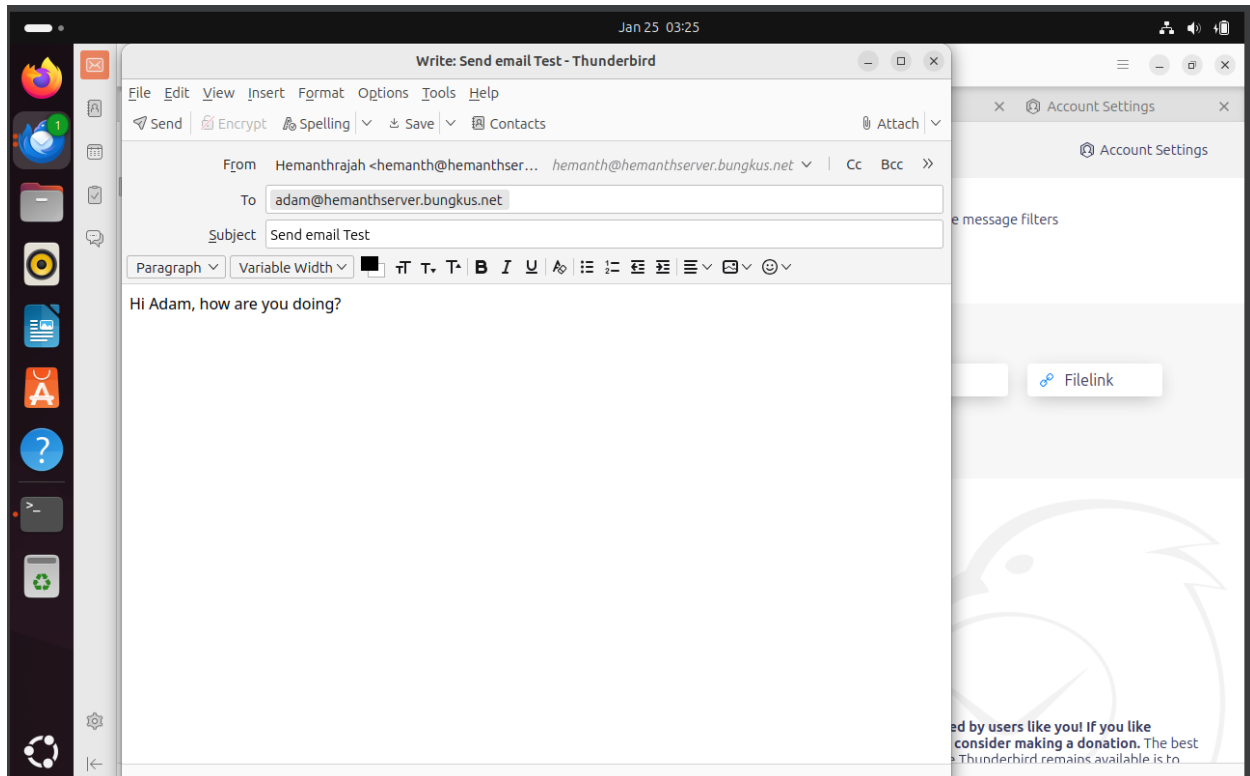


Figure 42: Sending email in Thunderbird

Sending the Email: I clicked Write in Thunderbird, entered the recipient's email address, and composed a simple message. After clicking Send, Thunderbird attempted to connect to the SMTP server.

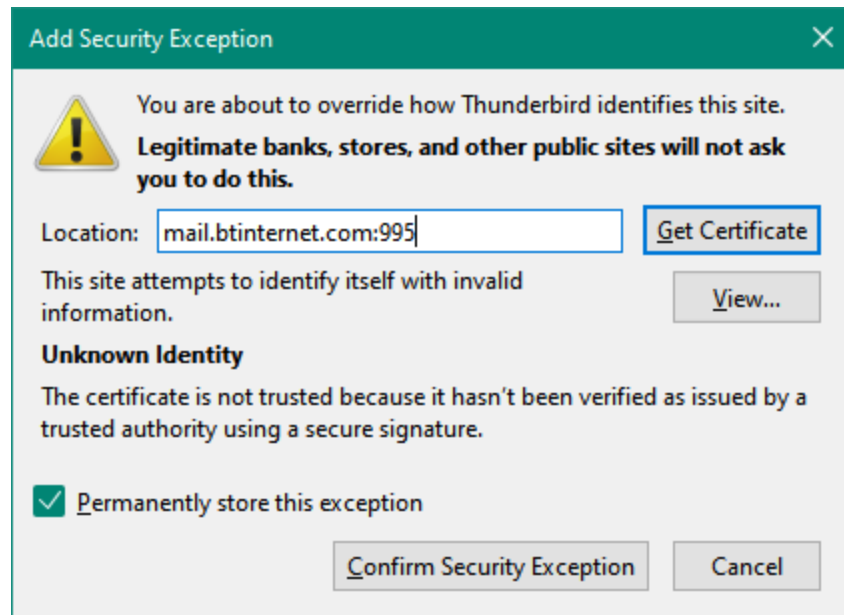


Figure 43: Confirm Security Exception (ColinT_Prest, 2024)

SSL/TLS Certificate Warning: Since I was using a self-signed SSL certificate, Thunderbird displayed a warning about the certificate being untrusted. To resolve this, I manually accepted the certificate by clicking **Accept the Risk and Continue**. This action added the server's certificate to Thunderbird's trusted certificate store. (Picture shown above is from the internet as I didn't screenshot mine when it appeared)

5.2.3 Testing Email Receiving Functionality

To test the ability to receive emails, I sent an email from hemanth@hemanthserver.bungkus.net to another email from my domain which was adam@hemanthserver.bungkus.net. I then checked Thunderbird for new messages.

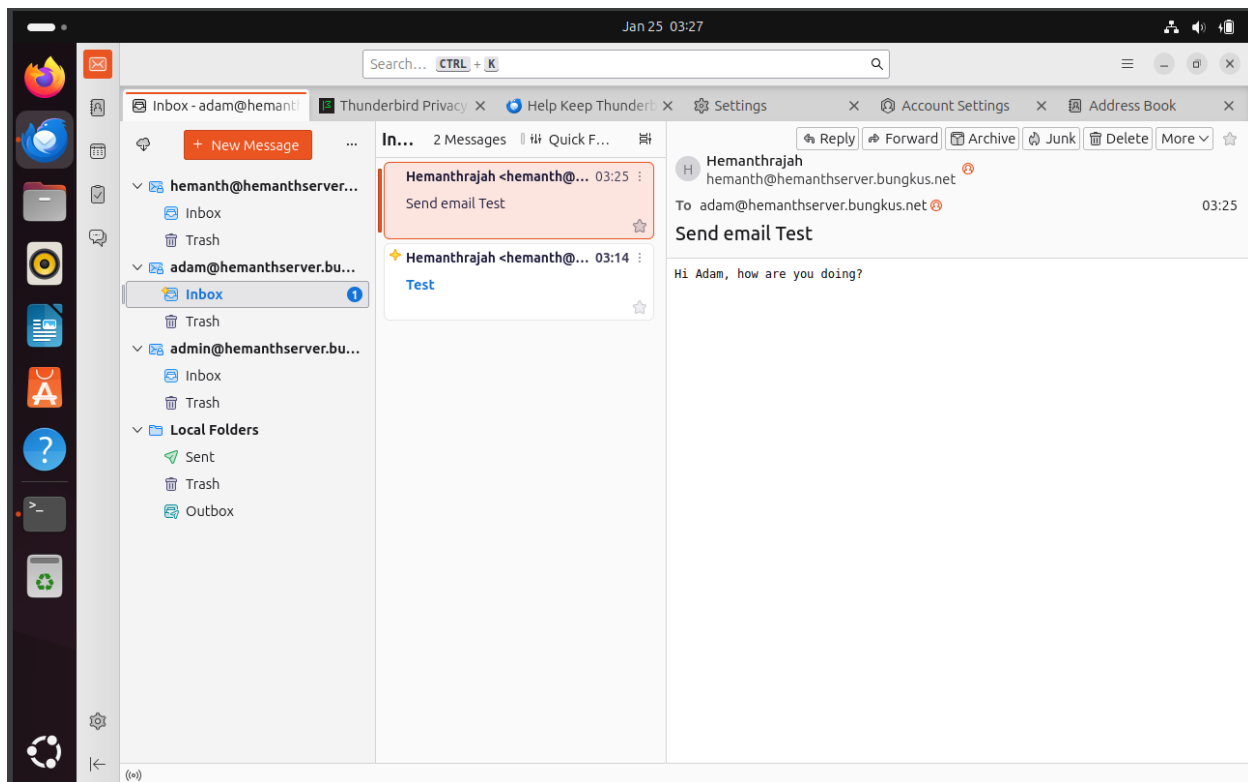


Figure 44: Receiving email in Thunderbird

Checking for New Emails: I clicked Get Messages in Thunderbird to retrieve any emails from the server. Thunderbird connected to the IMAP server on port 993 and attempted to download the emails.

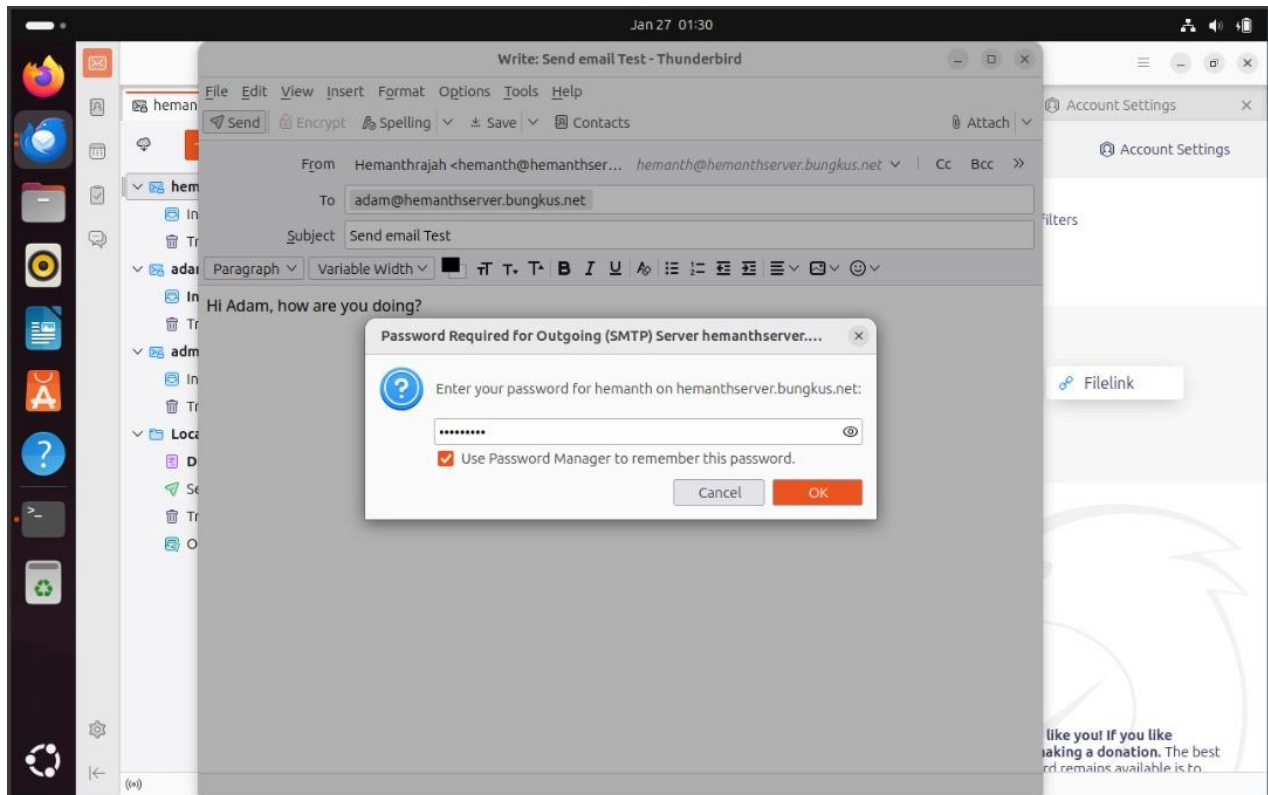


Figure 45: Password Required

Authentication Prompt: During the initial connection, Thunderbird prompted me for the email account password. After entering the correct password, I ensured that Thunderbird remembered it for future use.

Successful Email Retrieval: After accepting the certificate, Thunderbird successfully retrieved the email from the server. The test email appeared in the inbox, confirming that the IMAP server was functioning as expected. This is proven in Figure 44.

5.3.4 Troubleshooting Common Issues

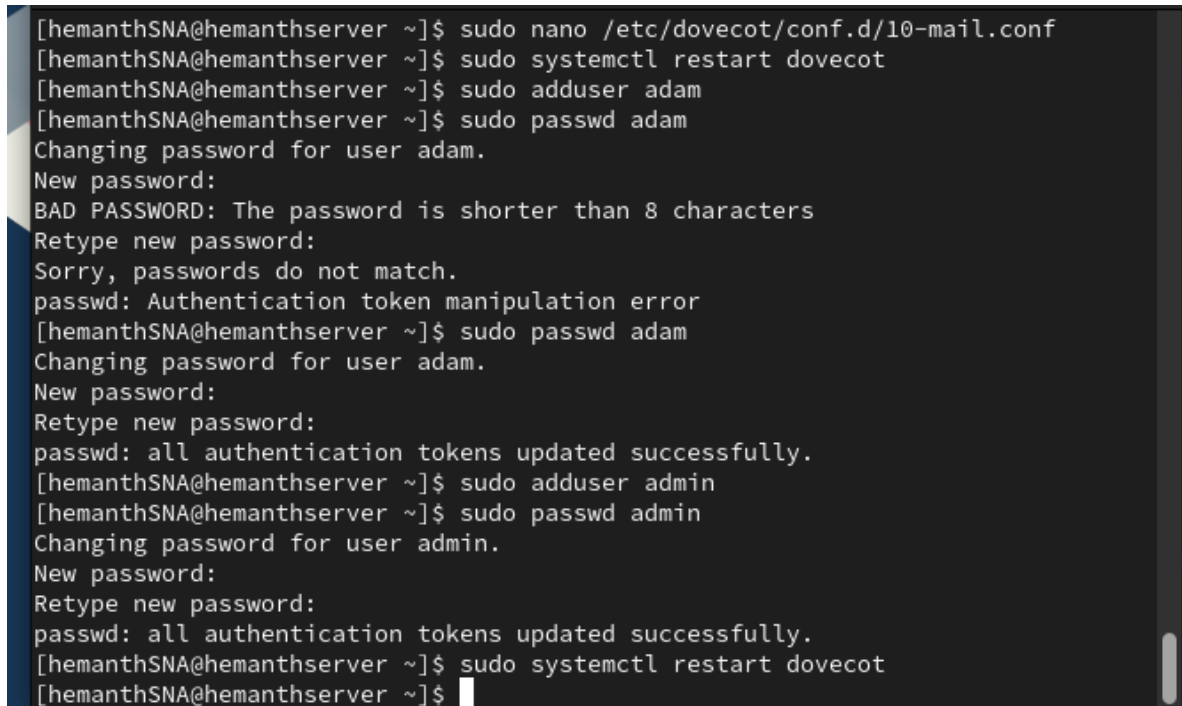
During testing, I encountered a few challenges, which I resolved through troubleshooting:

1. SSL/TLS Connection Errors:

1.1. **Issue:** Thunderbird displayed warnings about the SSL/TLS certificate.

1.2. **Resolution:** I manually imported and trusted the self-signed certificate in Thunderbird.

2. Login Failures:

A terminal window screenshot showing a series of commands and their outputs. The user is hemanthSNA@hemanthserver. The commands executed are: 'sudo nano /etc/dovecot/conf.d/10-mail.conf', 'sudo systemctl restart dovecot', 'sudo adduser adam', and 'sudo passwd adam'. The 'passwd' command for 'adam' initially fails with the message 'BAD PASSWORD: The password is shorter than 8 characters'. After retyping a longer password, it succeeds with 'passwd: Authentication token manipulation error'. Then, 'sudo adduser admin' and 'sudo passwd admin' are executed, with the password change for 'admin' also succeeding after a successful retype. Finally, 'sudo systemctl restart dovecot' is run. The terminal ends with a prompt for hemanthSNA@hemanthserver.

```
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-mail.conf
[hemanthSNA@hemanthserver ~]$ sudo systemctl restart dovecot
[hemanthSNA@hemanthserver ~]$ sudo adduser adam
[hemanthSNA@hemanthserver ~]$ sudo passwd adam
Changing password for user adam.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
[hemanthSNA@hemanthserver ~]$ sudo passwd adam
Changing password for user adam.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[hemanthSNA@hemanthserver ~]$ sudo adduser admin
[hemanthSNA@hemanthserver ~]$ sudo passwd admin
Changing password for user admin.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[hemanthSNA@hemanthserver ~]$ sudo systemctl restart dovecot
[hemanthSNA@hemanthserver ~]$
```

Figure 46: Change Password of Accounts

2.1. **Issue:** Thunderbird initially failed to authenticate the account.

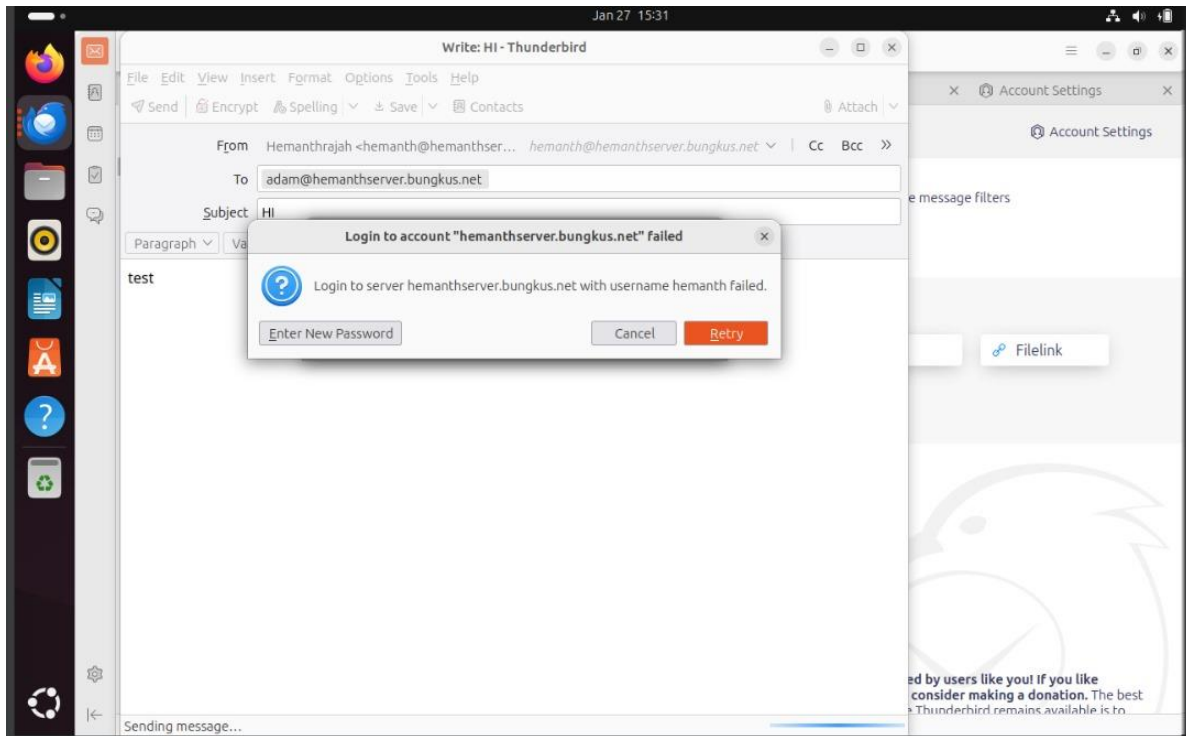


Figure 47: Login Failed in Thunderbird

2.2. Resolution: I ensured that the correct username (full email address) and password were entered. I also verified this by changing the passwords for my main user and creating new accounts in Rockey Server to be logged in to Thunderbird. I Then checked if the username and the details related to the account matches.

```
[hemanthSNA@hemanthserver ~]$ sudo nano /etc/dovecot/conf.d/10-mail.conf
[hemanthSNA@hemanthserver ~]$ sudo doveadm user Hemanth
field      value
user       hemanth
uid         1001
gid         1001
home        /home/hemanth
mail        maildir:/home/hemanth/Maildir
system_groups_user      hemanth
[hemanthSNA@hemanthserver ~]$
```

Figure 48: sudo doveadm user

3. Unable to Connect to the Server:

3.1. **Issue:** Thunderbird could not connect to the server on specified ports.

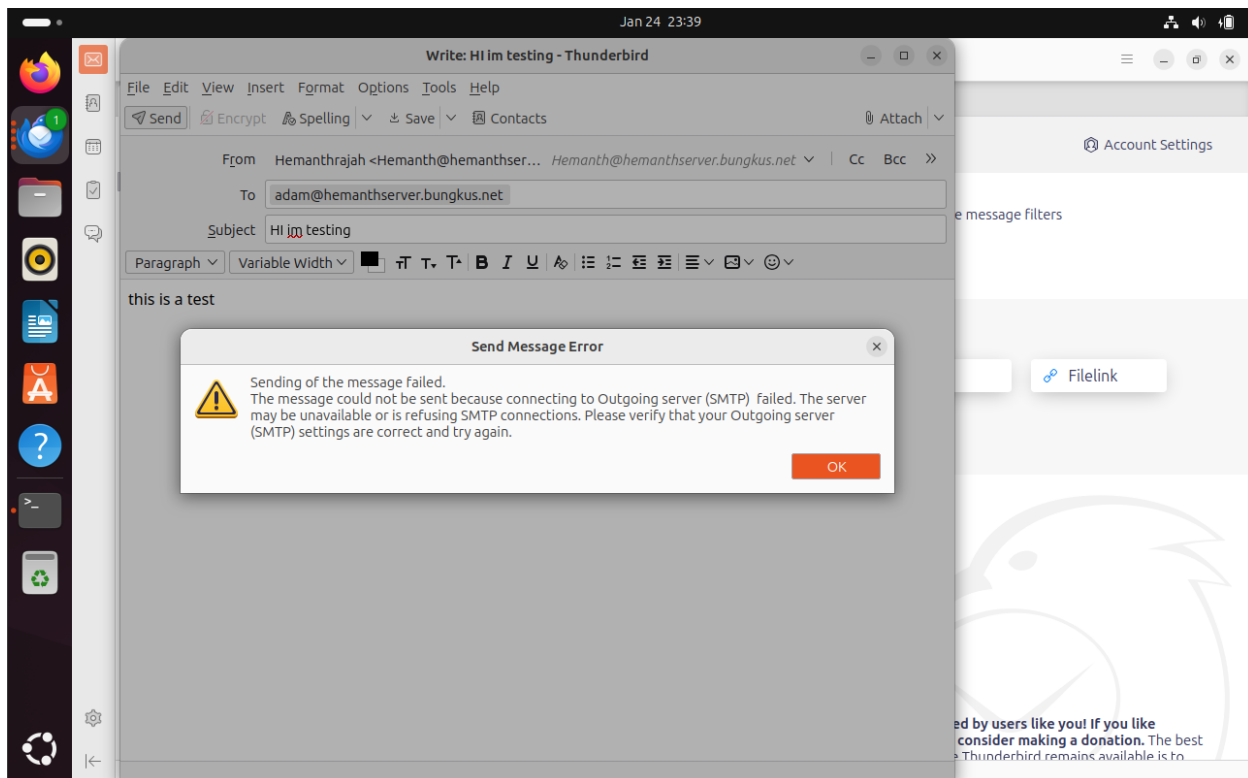


Figure 49: Send Message Error

3.2. Resolution: I verified that the required ports (993 for IMAP and 465 for SMTP) were open on the server firewall and checked if the ports were open in /etc/postfix/master.cf

```
GNU nano 5.6.1 /etc/postfix/master.cf
#
# Postfix master process configuration file. For details on the format
# of the file, see the master(5) manual page (command: "man 5 master" or
# on-line: http://www.postfix.org/master.5.html).
#
# Do not forget to execute "postfix reload" after editing this file.
#
# =====
# service type private unpriv chroot wakeup maxproc command + args
# (yes) (yes) (no) (never) (100)
# =====
smtp inet n - n - - smtpd
#smtp inet n - n - 1 postscreen
#smtpd pass - - n - - smtpd
#dnsblog unix - - n - 0 dnsblog
#tlsproxy unix - - n - 0 tlsproxy
submission inet n - n - - smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
# -o smtpd_tls_auth_only=yes
# -o smtpd_reject_unlisted_recipient=no
# -o smtpd_client_restrictions=permit_sasl_authenticated,reject
# -o smtpd_helo_restrictions=$mua_helo_restrictions
# -o smtpd_sender_restrictions=$mua_sender_restrictions
# -o smtpd_recipient_restrictions=
# -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
# -o milter_macro_daemon_name=ORIGINATING
smtps inet n - n - - smtpd
-o syslog_name=postfix/smtps
-o smtpd_tls_wrappermode=yes
-o smtpd_sasl_auth_enable=yes
# -o smtpd_reject_unlisted_recipient=no
# -o smtpd_client_restrictions=permit_sasl_authenticated,reject
# -o smtpd_helo_restrictions=$mua_helo_restrictions
# -o smtpd_sender_restrictions=$mua_sender_restrictions
# -o smtpd_recipient_restrictions=
# -o smtpd_relay_restrictions=permit_sasl_authenticated,reject
# -o milter_macro_daemon_name=ORIGINATING
#628 inet n - n - - qmqpd
pickup unix n - n 60 1 pickup
cleanup unix n - n - 0 cleanup
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location ^M-U Undo ^M-A Set Mark ^M-] To Bracket
^X Exit ^R Read File ^L Replace ^U Paste ^J Justify ^_ Go To Line ^M-E Redo ^M-G Copy ^M-^ Where Was
```

Figure 50: sudo nano /etc/postfix/master.cf

My Postfix was only listening on port 25, which is the default port for sending mail (SMTP). For secure email communication, Postfix should also listen on port 465 (SMTPS) for encrypted communication and optionally on port 587 (submission) for sending emails securely. After enabling the submission (Port 587) and smtps (Port 465), I also enabled firewalls on all ports which I was using which are 993 for IMAP, 995 for POP3, 465 for SLS/TLS and 587 for STARTTLS. Note that the testing of sending/receiving emails in Thunderbird for incoming server port was 993 and as for the outgoing server port, it was port 465.

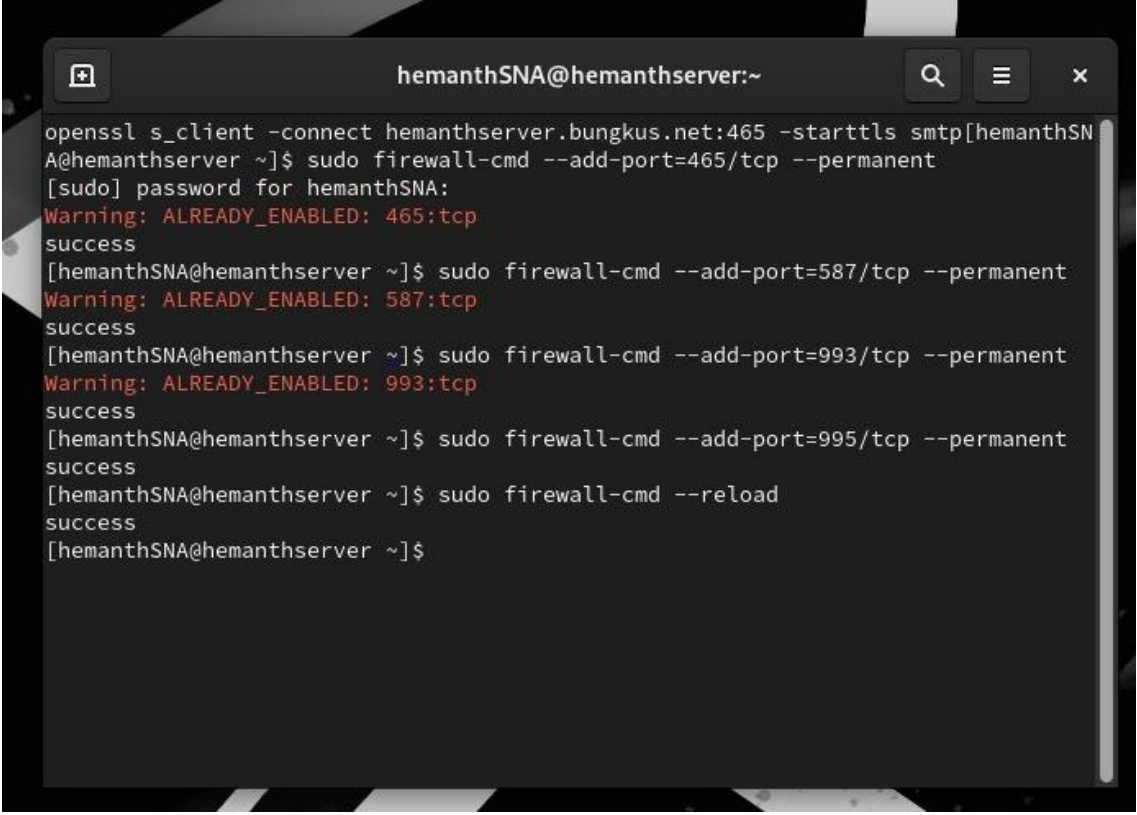
A terminal window titled 'hemanthSNA@hemanthserver:~' with search, menu, and close buttons. The terminal shows the execution of several commands to configure the firewall. The first command is 'openssl s_client -connect hemanthserver.bungkus.net:465 -starttls smtp[hemanthSNA@hemanthserver ~]\$ sudo firewall-cmd --add-port=465/tcp --permanent', which prompts for a password and shows a warning 'Warning: ALREADY_ENABLED: 465:tcp' followed by 'success'. Subsequent commands add ports 587/tcp, 993/tcp, and 995/tcp, each with the same warning and success message. The final command is 'sudo firewall-cmd --reload', which also shows a success message. The prompt returns to '[hemanthSNA@hemanthserver ~]\$'.

Figure 51: Enabling Firewall

Testing the email server with Thunderbird was a crucial step in ensuring the functionality and reliability of my setup. By carefully configuring the email client and addressing issues such as SSL/TLS warnings and login errors, I successfully verified that the server could send and receive emails securely (Wells, 2024). Monitoring the server logs provided additional confirmation that all components in Postfix, Dovecot, and SSL/TLS were working harmoniously. This process not only validated my configuration but also reinforced the importance of thorough testing and troubleshooting in email server deployments.

6.0 Conclusion

Using Ubuntu as the client and Rocky Linux as the server operating system, this project has been a thorough and hands-on exploration of the setup and deployment of an email server. Utilizing Oracle VirtualBox to establish a virtualized environment enabled me to replicate a real-world configuration, including client-server interactions. Throughout this process, I received important expertise in setting up, configuring, and debugging major components of an email server, including Postfix, Dovecot, and SSL/TLS encryption.

A crucial element of this task involved comprehending and establishing Postfix as the Mail Transfer Agent (MTA). Postfix functioned as the foundation for managing the transmission, reception, and forwarding of emails. By carefully changing the main.cf file, I adjusted Postfix to fit my domain, bungkus.net, and server FQDN, hemanthserver.bungkus.net. Configurations, including the definition of trusted networks, establishment of the origin domain, and activation of secure SMTP connections, ensured that my email server conformed to industry standards while being tailored to my individual context. This step enhanced my understanding of the essential function Postfix serves in regulating email traffic and preserving the integrity of email interactions.

The subsequent essential element was Dovecot, which I installed to facilitate IMAP and POP3 services for email retrieval. This procedure entailed specifying mail storage formats, facilitating user authentication, and integrating Dovecot with Postfix for cohesive functionality. Configuring Dovecot to utilize the Maildir format and activating SSL/TLS for secure email retrieval enhanced both the security and functionality of my server. I also evaluated Dovecot's authentication processes to ensure that user credentials were handled securely, which highlighted the importance of user privacy in email systems.

Security was a primary focus throughout this assignment. By generating and setting SSL/TLS certificates, I assured that all communication between the client and server was encrypted. Despite

utilizing self-signed certificates for this configuration, the procedure illustrated how encryption protects sensitive information, including user credentials and email content, from possible interception. Configuring Postfix and Dovecot to utilize these certificates greatly enhanced my understanding of secure communication methods in email servers.

The final stage of this project entailed evaluating the functionality of the email server using Thunderbird as the client program. Configuring Thunderbird to connect to my server requires great attention to detail, especially when manually inputting server information such as ports, encryption methods, and authentication credentials. Testing email capabilities using Thunderbird gave a practical approach to verify that the setups I had implemented were operating as planned. I encountered problems throughout this stage, such as certificate warnings and login errors, but tackling these issues through troubleshooting enhanced my problem-solving skills and deepened my grasp of email protocols.

Through this assignment, I also learned the value of system logs in monitoring and debugging server activities. By monitoring logs from Postfix and Dovecot, I was able to trace email flows, identify potential difficulties, and confirm effective operations. This incident underlined the crucial role of log files in managing and troubleshooting email servers.

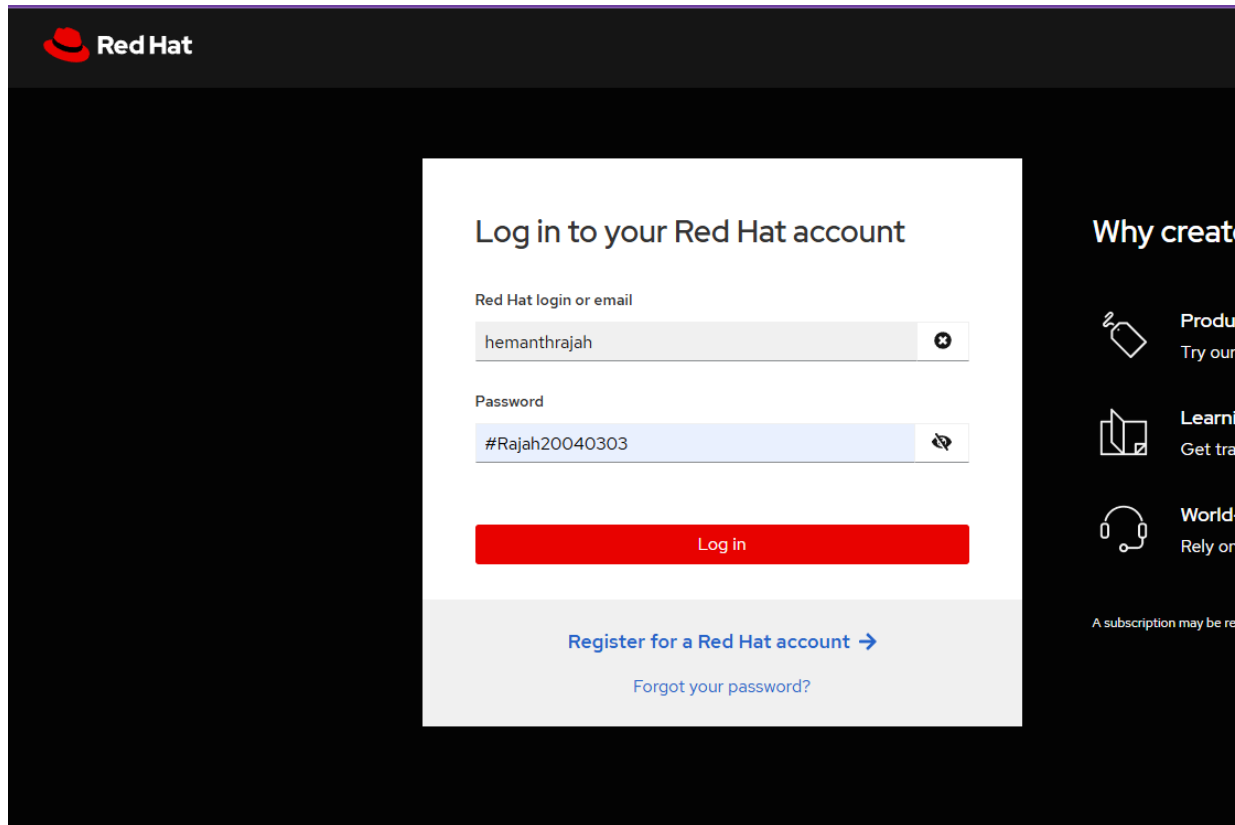
Working in a virtualized environment utilizing Oracle VirtualBox was another key part of this project. Virtualization provided a versatile and separated testing platform, enabling me to experiment with setups without harming other systems. Setting up static IPs, setting DNS, and verifying that the client and server could communicate within the virtual network allowed me to imitate a real-world scenario efficiently.

Overall, this assignment has been a wonderful learning experience, bringing insights into the nuances of email server setup and operation. It required me to use theoretical knowledge in a real situation, strengthening my skills in Linux server administration, network configuration, and

security measures. By documenting each step methodically, I have built a resource that can serve as a reference for future projects or comparable deployments.

In conclusion, establishing an email server from scratch using Rocky Linux and Ubuntu has given me a tremendous understanding for the intricacy and relevance of email systems in today's digital world. From handling email flows using Postfix to safeguarding connections with SSL/TLS, every step of this procedure has contributed to my growth as a system administrator. This project has not only enhanced my technical skills but also equipped me to confront real-world issues in designing and administering secure email servers.

7.0 Red Hat Academy Course Completion



The image shows the Red Hat Academy login page. At the top left is the Red Hat logo. The main heading is "Log in to your Red Hat account". Below this are two input fields: "Red Hat login or email" with the value "hemanthrajah" and "Password" with the value "#Rajah20040303". A red "Log in" button is positioned below the password field. Underneath the button is a link "Register for a Red Hat account →" and a link "Forgot your password?". To the right of the login form, there is a section titled "Why create" with three icons and text: a tag icon for "Produce", a book icon for "Learn", and a headset icon for "World". At the bottom right, there is a small text "A subscription may be re".

Red Hat

Log in to your Red Hat account

Red Hat login or email

hemanthrajah

Password

#Rajah20040303

Log in

[Register for a Red Hat account →](#)

[Forgot your password?](#)

Why create

Produce
Try our

Learn
Get tra

World
Rely on

A subscription may be re

Figure 52: Red Hat Academy Login

Username: Hemanthrajah

Password: Rajah20040303

Personal info

Manage your basic information

General

Jump to section

Select a locale	English ▼
Username *	hemanthrajah
Email *	tp070844@mail.apu.edu.my
First name	HEMANTHRAJAH
Last name	KRISHNARAJAH

Save

Cancel

General

Figure 53: Red Hat Academy information

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